

Determinacy and Modality in Positive Characteristic

Yotam Svoray

Abstract

We study finite determinacy and modality in positive characteristic through three complementary constructions: prescribed-jet transversality, Frobenius deformation directions, and a finite flat critical cover. For matrix germs over an arbitrary field, prescribed-jet transversality to the rank stratification shows that finite left-right determinacy is equivalent to finite codimension of the tangent image. Analogous results hold for pure right equivalence, for one-sided actions over infinite fields, and in specified ranges over finite fields. These methods also yield a characterization of finite contact determinacy for positive-dimensional ideals and imply expected-height, Cohen–Macaulayness, reducedness, and normality results for determinantal ideals. For isolated hypersurface singularities over an algebraically closed field of characteristic $p > 0$, we introduce a Frobenius algebra whose length and Loewy structure govern the Milnor number, determinacy, and essential corank. Using a finite flat critical cover, we prove that proper modality coincides with right modality. It follows that, in fixed characteristic, bounded right modality gives an ambient-dimension-independent bound on determinacy and, up to nonsingular quadratic suspension, a finite algebraic parametrization by modular families. The same Frobenius methods yield uniform finiteness results for F -jumping numbers and show that the F -pure threshold is determined at a single Frobenius level.

1 Introduction

Finite determinacy is one of the main tools in the study and classification of singularities. It allows one to replace a formal power series, or a matrix of power series, by a finite jet without changing its equivalence class. Over a field of characteristic zero, finite determinacy is closely related to the tangent space of the corresponding orbit. In positive characteristic the relation is subtler, since an orbit map need not be separable and its tangent image can be strictly smaller than the tangent space to the orbit; see [16, 13].

Understanding this subtly lets us approach this problem from a common principle. Specifically, finite determinacy allows geometric information obtained from sufficiently high-order perturbations to be transported back to the original germ. For matrix germs, the perturbations are chosen by prescribed-jet transversality. For hypersurface germs, Frobenius supplies deformation directions that survive modulo the infinitesimal right action, while the critical-cover construction converts Milnor numbers into local degrees of a finite map. These mechanisms lead, respectively, to tangent-image criteria, uniform bounds at fixed modality, and the comparison of proper and right modality. We state the structural

results first and then record the F -singularity consequences.

Consequently, we begin with matrix germs over an arbitrary field. Greuel–Pham proved that finite codimension of the appropriate tangent image implies finite determinacy over an arbitrary field, and proved the converse when the relevant orbit maps are eventually separable [16]. The converse over arbitrary fields is stated as Conjecture 3.1.16 in [13]. Our prescribed-jet approximation theorem, Theorem 2.12, applies to maps into a variety with a finite smooth stratification. Over finite fields we use Poonen’s closed-point sieve with prescribed Taylor data [35], imposing simultaneously the first-jet smoothness conditions required on the finitely many strata; over infinite fields the corresponding statement follows from an incidence-dimension argument. Applied to the rank stratification, it gives Theorem 2.13. The one-sided actions are then reduced to kernel and image maps to Grassmannians by Proposition 2.20, and the remaining punctured critical loci are eliminated by the Frobenius gauges of Lemma 2.25. The pure right action is treated separately in Theorem 2.32.

Theorem 1.1. *Let K be a field, put $R = K[[x_1, \dots, x_s]]$, $\mathfrak{m} = \langle x_1, \dots, x_s \rangle$, and $M_{m,n} = \text{Mat}(m, n, R)$. We write $\mathcal{R} = \text{Aut}_K(R)$, $\mathcal{G}_l = \text{GL}_m(R) \rtimes \mathcal{R}$, $\mathcal{G}_r = \text{GL}_n(R)^{\text{op}} \rtimes \mathcal{R}$, and $\mathcal{G}_{lr} = (\text{GL}_m(R) \times \text{GL}_n(R)^{\text{op}}) \rtimes \mathcal{R}$. For $1 \leq t \leq \min(m, n)$, let $I_t(A)$ be the ideal generated by the $t \times t$ minors of A and put $c_t = (m - t + 1)(n - t + 1)$. Then:*

- (i) *For $G \in \{\mathcal{R}, \mathcal{G}_{lr}\}$, finite G -determinacy is equivalent to finite codimension of the G -tangent image. If K is infinite, the same equivalence holds for $G \in \{\mathcal{G}_l, \mathcal{G}_r\}$. Over an arbitrary field, the one-sided equivalence also holds when $n \leq m$ and $s \leq m - n + 1$ for $\mathcal{G}_l$, and when $m \leq n$ and $s \leq n - m + 1$ for $\mathcal{G}_r$; see Theorems 2.15, 2.27, 2.23, and 2.32.*
- (ii) *A proper ideal $I \subseteq R$ with $\dim\left(\frac{R}{I}\right) > 0$ is finitely contact determined if and only if $\frac{R}{I}$ is an isolated complete-intersection singularity; see Corollary 2.10.*
- (iii) *If A is finitely determined for one of the actions $\mathcal{R}, \mathcal{G}_l, \mathcal{G}_r, \mathcal{G}_{lr}$, then, for every $1 \leq t \leq \min\{m, n\}$ we have $\text{ht } I_t(A) = \min\{s, (m - t + 1)(n - t + 1)\}$; see Theorem 2.37.*
- (iv) *Under the hypotheses of (iii), the quotient $\frac{R}{I_t(A)}$ is Cohen–Macaulay. Moreover, it is reduced if $(m - t + 1)(n - t + 1) < s$, and it is a normal domain if $(m - t + 1)(n - t + 1) \leq s - 2$; see Theorem 2.37.*

Keeping these distinctions in view, Theorem 1.1 gives the tangent-image criterion for the full left–right and pure right actions over every field, and for all four actions over every infinite field. Over finite fields, the one-sided criteria are established in the dimension ranges stated in Assertion (i).

Yet another setting is that of isolated hypersurface singularities over an algebraically closed field k of characteristic $p > 0$. The Frobenius algebra and Frobenius modality are defined in Definition 3.15. Proposition 3.17 shows that the Frobenius directions give genuine moduli. Their length and Loewy structure then control the Milnor number, separability of high jet orbits, determinacy, multiplicity, and essential corank; see Theorems 3.19, 3.11,

and 3.25.

Another main input is the splitting lemma recorded in Lemma 3.4(vii). We write $c(f)$ for the *essential corank*, namely the number of variables in its residual germ after a nonsingular quadratic summand has been split off. Finally, the critical-cover argument of Section 4 identifies proper modality with right modality in Theorem 4.16.

Theorem 1.2 (Modality and uniform determinacy). *Let k be algebraically closed of characteristic $p > 0$, and let f be an isolated hypersurface germ.*

- (i) *We have $\text{Fmod}(f) \leq \text{pmod}(f) = \text{Rmod}(f)$; see Proposition 3.17 and Theorem 4.16.*
- (ii) *We have $\mu(f) + c(f) \leq p^{c(f)}(\text{Fmod}(f) + 1)$. Consequently, we have that $\mu(f) \leq p^{2 + \sqrt[3]{6 \text{Rmod}(f)}}(\text{Rmod}(f) + 1)$; see Corollary 4.5.*
- (iii) *For every $D \geq 0$, the residual number of variables, Milnor number, multiplicity, and right-classification radius of all isolated germs with $\text{Rmod}(f) \leq D$ are bounded in terms of p and D . Moreover, there is an integer $\hat{N}_{p,D}$, independent of the original ambient dimension, such that every such f is right $\hat{N}_{p,D}$ -determined; see Theorem 4.3 and Corollary 4.4.*

Leveraging a Rosenlicht stratification of those jet spaces and a subsequent finite-multisection argument yields a finite algebraic parametrization. The relevant finite-multisection principle is Lemma 5.2, and stable cancellation is Lemma 5.3.

Theorem 1.3 (Finite modular classification). *Fix $p > 0$ and $D \geq 0$. Up to nonsingular quadratic suspension, the isolated germs f with $\text{Rmod}(f) \leq D$ are covered by finitely many algebraic families $F_i(x, t)$, with $t \in T_i$, such that:*

- (i) *$\dim T_i \leq D$, the number of residual variables is bounded in terms of p, D , and the degrees of the representatives are bounded in terms of p, D ;*
- (ii) *every stable right-equivalence class occurs;*
- (iii) *each stable class occurs at only finitely many points of each T_i .*

The families may be refined so that Milnor number and right modality are constant.

Lastly, we apply the bounded polynomial representatives to F -singularity invariants. The denominator theorem of Blickle–Mustață–Smith then places all F -jumping numbers in $(0, 1]$ in a finite set depending only on p and the modality bound. The resulting uniform separation of possible thresholds implies that one sufficiently high Frobenius level determines the F -pure threshold. The remaining applications use the results of Katzman–Lyubeznik–Zhang and Jiang [25, 24] on test ideals, and of Blickle–Stäbler [5] on positive-characteristic Bernstein–Sato roots.

Theorem 1.4 (F -singularity consequences). *Fix $p > 0$ and $D \geq 0$, and let f be isolated with $\text{Rmod}(f) \leq D$.*

- (i) Every F -jumping number of $\langle f \rangle$ in $(0, 1]$ belongs to a finite set $\Xi_{p,D}$ depending only on p, D ; see Theorem 6.7.
- (ii) There is an integer $E_{p,D}$ such that the single Frobenius-level invariant $\nu_{E_{p,D}}(f)$ determines $\text{fpt}(f)$; see Theorem 6.9.
- (iii) If $\frac{R}{\langle f \rangle}$ is reduced, then the number of F -jumping coefficients in $[0, 1]$ and the colength of its finitistic test ideal are bounded in terms of p, D . Moreover, a power of the maximal ideal, with exponent depending only on p, D , is contained in the Jacobian ideal and hence in the finitistic test ideal; see Corollary 6.11.
- (iv) Only finitely many eventual sequences of level- e positive-characteristic Bernstein–Sato root sets occur at fixed p, D ; Corollary 6.12.

In addition, the numerical F -jumping spectrum is uniformly $\mathfrak{m}$ -adically constant in the sense of Corollary 6.5.

Organization. In Section 2 we study matrix determinacy and prove the prescribed-jet approximation results. In Section 3 we study high-jet stabilizers and define the Frobenius algebra. In Section 4 we use Lemma 3.4(vii) and the critical cover to obtain uniform bounds and to prove equality of proper and right modality. In Section 5 we prove the finite algebraic parametrization theorem. Finally, in Section 6 we give the applications to F -singularity invariants.

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2 Matrix determinacy over arbitrary fields

In this section, we study finite determinacy of matrices over an arbitrary field. We first recall the tangent-image criterion of Greuel–Pham and the two questions from [13] that we address. We then prove the prescribed-jet approximation theorem and apply it to the full left–right action, to one-sided actions, and to determinantal ideals.

Retain the notation $K, m, n, s, R, \mathfrak{m}, M_{m,n}, \mathcal{R}, \mathcal{G}_l, \mathcal{G}_r, \mathcal{G}_{lr}$ from the Introduction. Put $P = K[x_1, \dots, x_s]$, $\mathfrak{m}_0 = \langle x_1, \dots, x_s \rangle \subseteq P$, and $S_0 = P_{\mathfrak{m}_0}$. Thus R is the $\mathfrak{m}_0$ -adic completion of S_0 and $\mathfrak{m} = \mathfrak{m}_0 R$. We write $M_m(R) = M_{m,m}(R)$, $\text{ht}(I)$ for the height of an ideal I , and $\kappa(\mathfrak{p})$ for the residue field of $R_{\mathfrak{p}}$. The notation $\text{Gr}(r, N)$ denotes the Grassmannian of r -dimensional subspaces of an N -dimensional vector space. If M is a finite-length module over a local ring A , then $\text{length}_A(M)$ denotes its module length. For a vector-valued germ $H = (H_1, \dots, H_q)$, we write $\text{JacMat}(H) = \left(\frac{\partial H_i}{\partial x_j} \right)_{i,j}$ for its Jacobian matrix. For $\ell \geq 0$ put $R_\ell = \frac{R}{\mathfrak{m}^{\ell+1}}$ and $D_\ell = \dim_K R_\ell = \binom{s+\ell}{s}$. We use multi-index notation throughout. If $\alpha = (\alpha_1, \dots, \alpha_s) \in \mathbb{N}^s$, then $|\alpha| = \sum_i \alpha_i$ and $x^\alpha = x_1^{\alpha_1} \cdots x_s^{\alpha_s}$. For $\beta \leq \alpha$, the inequality is coordinatewise and $\binom{\alpha}{\beta} = \prod_i \binom{\alpha_i}{\beta_i}$. The symbol p is reserved for a prime number and, in

the hypersurface part, always denotes the characteristic.

We use the four matrix actions $\mathcal{R}, \mathcal{G}_l, \mathcal{G}_r, \mathcal{G}_{lr}$ introduced in the Introduction; these are the conventions of Greuel–Lossen–Shustin [13, §3.1.3]. For $G \in \{\mathcal{R}, \mathcal{G}_l, \mathcal{G}_r, \mathcal{G}_{lr}\}$ and $k \geq 0$, the k -jet of $A \in \mathfrak{m}M_{m,n}$ is its class $j^k A$ modulo $\mathfrak{m}^{k+1}M_{m,n}$. The matrix A is k - G -determined if every $B \in \mathfrak{m}M_{m,n}$ with $j^k B = j^k A$ is G -equivalent to A ; it is *finitely G -determined* if this holds for some k .

For an algebraic jet-group action, the *tangent image* at a jet is the image of the differential at the identity of the orbit map. Passing back to formal power series gives the tangent-image modules used here. For example, the extended left–right tangent image is $\hat{T}_A^e(\mathcal{G}_{lr}A) = M_m(R)A + AM_n(R) + \sum_{\nu=1}^s R \frac{\partial A}{\partial x_\nu}$, while the ordinary tangent image has $\mathfrak{m}$ instead of R in front of the derivative terms. The analogous formulas are obtained by omitting the right, left, or both matrix terms for $\mathcal{G}_l, \mathcal{G}_r, \mathcal{R}$, respectively; see [13, Definition 3.1.11] and [16, §2].

Theorem 3.1.12 of [13], originating in [16], proves that finite codimension of the appropriate tangent image is sufficient for finite determinacy over an arbitrary field. The general problem is the necessity of this condition. For the pure right and full left–right actions, we prove necessity over an arbitrary field; see Theorems 2.32 and 2.15. For the one-sided actions, we prove necessity over every infinite field in Theorem 2.27. Over an arbitrary field, including a finite field, we obtain the one-sided equivalence in the dimension ranges of Theorem 2.23. The one-sided argument over infinite fields reduces the problem to critical loci of the kernel and image maps to Grassmannians and eliminates these loci by combining deep-gauge general position with Frobenius gauges.

Question A (GLS Conjecture 3.1.16). Is the tangent-image criterion also necessary for finite determinacy over every field for $G \in \{\mathcal{R}, \mathcal{G}_l, \mathcal{G}_r, \mathcal{G}_{lr}\}$?

The book records this as open in positive characteristic; see [13, Conjecture 3.1.16 and Remark 3.1.16.1]. It is known when the relevant orbit maps are eventually separable, in particular in characteristic zero; see [13, Theorem 3.1.21] and [16, Theorem 4.3]. It is also known in several special cases, including scalar functions and one-column left–right matrices under the hypotheses stated in [13, Theorems 3.1.9 and 3.1.23]. Theorems 2.15 and 2.32 give an affirmative answer for the full left–right and pure right actions over every field. For the one-sided actions, Theorem 2.27 gives an affirmative answer over every infinite field, while Theorem 2.23 gives the corresponding arbitrary-field statement in the ranges $n \leq m$ and $s \leq m - n + 1$ for $\mathcal{G}_l$, and $m \leq n$ and $s \leq n - m + 1$ for $\mathcal{G}_r$.

Remark 2.1. Finite determinacy for smooth and analytic map germs goes back to the work of Mather on finitely determined map germs [28]. For hypersurface functions over fields of arbitrary characteristic, finite right determinacy is equivalent to finiteness of the Milnor number by Boubakri–Greuel–Markwig [6]. For matrices in positive characteristic, Greuel–Pham introduced the tangent image in order to separate the infinitesimal image of

the orbit map from the tangent space of a possibly inseparable orbit [16]. This distinction is the source of Question A. For determinacy and Mather–Yau-type phenomena over complete regular local rings in mixed characteristic, see [40].

For ideals, let $I = \langle h_1, \dots, h_r \rangle \subseteq R$ be minimally generated. Two ideals $I, J \subseteq R$ are *contact equivalent* if a K -algebra automorphism of R sends I onto J ; finite contact determinacy means finite determinacy for this equivalence, expressed through a minimal generator column.

Let $I = \langle h_1, \dots, h_r \rangle \subseteq R$ be minimally generated. We say that R/I is an *isolated complete-intersection singularity* (ICIS), if $h_1, \dots, h_r$ is a regular sequence and $\dim_K \left(\frac{R}{I + I_r(\text{JacMat}(h_1, \dots, h_r))} \right) < \infty$. Thus the Jacobian condition holds at every non-maximal prime of $\frac{R}{I}$. For algebraic representatives over a perfect ground field, this is equivalent to the usual punctured smoothness condition over K . Note that we also include the nonsingular case. Regard the generators as the column $A_I = (h_1, \dots, h_r)^\top \in M_{r,1}(R)$. Contact equivalence of such ideals agrees with left–right equivalence of their minimal generator columns; see [13, Proposition 3.1.26] and [17, Proposition 4.3].

Question B (GLS Problem 3.1.29). If K is finite and $\frac{R}{I}$ has positive dimension, does finite contact determinacy imply that $\frac{R}{I}$ is an ICIS?

The reverse implication and the equivalence between finite Tjurina number and the ICIS condition are already valid over arbitrary fields; the missing case recorded in [13, Problem 3.1.29] is precisely the finite-field implication from finite contact determinacy.

Lemma 2.2. *Let a group G act on an $\mathfrak{m}$ -adically filtered set E , and let P be a G -invariant property. Suppose that for every $A \in E$ and every N there exists $B \in A + \mathfrak{m}^N E$ satisfying P . Then every finitely G -determined element of E satisfies P .*

Proof. If A is k - G -determined, choose $B \in A + \mathfrak{m}^{k+1} E$ satisfying P . Then $A \sim_G B$, so $P(A)$ follows from G -invariance. $\square$

Proposition 2.3. *Let $G \in \{\mathcal{R}, \mathcal{G}_l, \mathcal{G}_r, \mathcal{G}_{lr}\}$ and let $A \in \mathfrak{m}M_{m,n}$ be finitely G -determined. Then $\text{rank}_{\text{Frac}(R)} A = \min(m, n)$.*

Proof. Put $r = \min(m, n)$ and suppose that A is k - G -determined. Fix an $r \times r$ submatrix A_0 of A , and choose $N \geq k + 1$. For $\lambda = (\lambda_1, \dots, \lambda_r) \in K^r$, perturb the diagonal of that block by $\lambda_1 x_1^N, \dots, \lambda_r x_1^N$ (with x_1 used repeatedly; no distinct variables are needed), leaving all other entries unchanged. Call the resulting matrix B_λ . Then $B_\lambda - A \in \mathfrak{m}^N M_{m,n}$. The determinant of the chosen block of B_λ is a polynomial $P(\lambda_1, \dots, \lambda_r) \in R[\lambda_1, \dots, \lambda_r]$ which has degree at most one in each λ_i and whose coefficient of $\lambda_1 \cdots \lambda_r$ is $x_1^{Nr} \neq 0$.

A nonzero polynomial of degree at most one in each variable cannot vanish on all of K^r : this follows by induction on r from the one-variable fact, valid over every field. Hence for some $\lambda \in K^r$ the chosen $r \times r$ minor of B_λ is nonzero. Therefore $\text{rank}_{\text{Frac}(R)} B_\lambda = r$. Finite determinacy gives $B_\lambda \sim_G A$. Generic rank is invariant under source automorphisms and under invertible row and column operations. Therefore A also has generic rank r . $\square$

Proposition 2.4. *Let $A \in \mathfrak{m}M_{m,n}$ be finitely determined. Then:*

- (i) *if A is finitely $\mathcal{R}$ -determined, then $mn \leq s$;*
- (ii) *if A is finitely $\mathcal{G}_l$ -determined, then $mn \leq m^2 + s$;*
- (iii) *if A is finitely $\mathcal{G}_r$ -determined, then $mn \leq n^2 + s$.*

In particular, when $n > m$, finite $\mathcal{G}_l$ -determinacy implies $m(n - m) \leq s$, and when $m > n$, finite $\mathcal{G}_r$ -determinacy implies $n(m - n) \leq s$.

Proof. Suppose A is k -determined and fix $\ell > k$. The set of ℓ -jets having the same k -jet as A is an affine cylinder of dimension $mn(D_\ell - D_k)$. For an infinite field, all its K -points lie in the ℓ -jet orbit of A , hence the Zariski closure of that orbit has dimension at least this number. The ℓ -jet automorphism group is the open subset of $\mathfrak{m}R_\ell^s$ on which the linear part is invertible. Hence its dimension is exactly $s(D_\ell - 1)$. Likewise, $\mathrm{GL}_m(R_\ell)$ and $\mathrm{GL}_n(R_\ell)$ are open subsets of $M_m(R_\ell)$ and $M_n(R_\ell)$, so their dimensions are exactly $m^2 D_\ell$ and $n^2 D_\ell$. Since orbit dimension is at most group dimension, division by D_ℓ followed by $\ell \rightarrow \infty$ gives the asserted inequalities.

If $K = \mathbb{F}_q$, we use cardinalities instead. The cylinder contains $q^{mn(D_\ell - D_k)}$ points. Reduction modulo $\mathfrak{m}$ gives

$$\begin{aligned} |\mathrm{Aut}_K(R_\ell)| &= |\mathrm{GL}_s(\mathbb{F}_q)| q^{s(D_\ell - 1 - s)}, \\ |\mathrm{GL}_m(R_\ell)| &= |\mathrm{GL}_m(\mathbb{F}_q)| q^{m^2(D_\ell - 1)}, \\ |\mathrm{GL}_n(R_\ell)| &= |\mathrm{GL}_n(\mathbb{F}_q)| q^{n^2(D_\ell - 1)}. \end{aligned}$$

The orbit has cardinality at most that of the acting jet group. Taking $\log_q$, dividing by D_ℓ , and passing to the limit gives the same bounds. $\square$

We next prove the approximation result over finite fields. We use the Bertini theorem of Poonen with Taylor conditions to keep a prescribed finite jet fixed while imposing smoothness away from the origin.

We first explain what we mean by prescribing Taylor data. If Z is a finite subscheme of an ambient affine or projective space, by *Taylor data on Z* we mean a prescribed section of $H^0(Z, \mathcal{O}_Z)$, namely the restriction of the defining polynomial or hypersurface section to Z . In the case used repeatedly below, $Z = \mathrm{Spec}\left(\frac{P}{\mathfrak{m}_0^N}\right)$ is the $(N - 1)$ -st infinitesimal neighborhood of the origin, and prescribing Taylor data is exactly prescribing a residue class modulo $\mathfrak{m}_0^N$, equivalently an $(N - 1)$ -jet. In multi-index notation, if $h = \sum_{\alpha \in \mathbb{N}^s} c_\alpha x^\alpha$, then this Taylor datum is represented by $\sum_{|\alpha| < N} c_\alpha x^\alpha$. Thus *Taylor data* here does not mean a choice of derivatives divided by factorials; it means the scheme-theoretic restriction to the chosen finite subscheme. This is the terminology used by Poonen in his Bertini theorem with Taylor conditions [35, Theorem 1.2].

Lemma 2.5. *Let $K = \mathbb{F}_q$. Fix a quasiprojective embedding of a K -scheme X into projective space, let Z be a finite subscheme of the ambient projective space, and suppose $U = X - (X \cap Z)$ is smooth of pure dimension $d > 0$. Given arbitrary prescribed Taylor data on Z , there exists, in sufficiently large degree, a hypersurface having exactly those Taylor data and whose intersection with U is smooth of pure dimension $d - 1$.*

Proof. This is the Bertini theorem of Poonen with Taylor conditions [35, Theorem 1.2]. If T is the singleton consisting of the prescribed restriction on Z , the density appearing in the theorem of Poonen is positive. Importantly, X need not be smooth along Z ; only $U = X - (X \cap Z)$ is required to be smooth. $\square$

For the stratified argument we use the following direct consequence of the infinitely-many-local-conditions theorem of Poonen [35, Theorem 1.3]. We record it explicitly because the simultaneous conditions are one of the main inputs in the proof of Theorem 2.12.

Lemma 2.6 (Local-conditions principle over finite fields). *Let $K = \mathbb{F}_q$, let Z be a finite subscheme of a projective space, and let $X_1, \dots, X_u$ be smooth quasiprojective subschemes. Fix Taylor data on Z . Then, in sufficiently large degree, there exists a hypersurface realizing those Taylor data such that, away from Z , its intersection with every positive-dimensional X_i is smooth of the expected dimension and it avoids every zero-dimensional X_i .*

Proof. Apply [35, Theorem 1.3]. At a closed point $P \notin Z$, simultaneous smoothness is a condition on the value and first derivatives of the hypersurface section. For each X_i containing P , the bad first jets are those for which the value and all tangent derivatives along X_i vanish. These are precisely the local smoothness conditions occurring in Poonen's closed-point sieve. The prescribed Taylor class changes only the finitely many local conditions supported on Z . Poonen's convergence estimates for the sieve therefore give a positive limiting density, and hence such hypersurfaces exist in sufficiently large degree. $\square$

Remark 2.7. Classical Bertini arguments over infinite fields obtain smooth members by removing a proper closed subset of a linear system. Over a finite field such an open set need not have a rational point. Poonen replaced this genericity argument by a closed-point sieve and proved positive-density Bertini theorems, including prescribed Taylor conditions [35]. Recent work of Bertucci develops further forms of Taylor conditions over finite fields [3]; Ajit–Bertucci prove complementary finite-field Bertini theorems preserving Hilbert–Samuel multiplicity [1].

Proposition 2.8. *Let $K = \mathbb{F}_q$ and let $r < s$. For arbitrary $h_1, \dots, h_r \in \mathfrak{m}$ and every $N \geq 1$, there exist $g_1, \dots, g_r \in \mathfrak{m}$ such that $g_i \equiv h_i \pmod{\mathfrak{m}^N}$ and $\frac{R}{(g_1, \dots, g_r)}$ is an isolated complete-intersection singularity.*

Proof. The successive construction in the proof of Theorem 2.9 uses no finite-determinacy assumption until its final transport step. Applying Lemma 2.5 successively produces a complete intersection that is smooth off the origin and realizes the prescribed jet. The height and Jacobian arguments given below then show that its completion is an ICIS. $\square$

Theorem 2.9. *Let $K = \mathbb{F}_q$, let $R = K[[x_1, \dots, x_s]]$, and let I be a proper ideal with $\dim(\frac{R}{I}) > 0$. If I is finitely contact determined, then $\frac{R}{I}$ is an isolated complete-intersection singularity.*

Proof. Choose a minimal system $I = \langle h_1, \dots, h_r \rangle$. We first show that $r < s$. The field-independent part of [13, Theorem 3.1.22(2)], equivalently [17, Theorem 2.6], implies that if a finitely determined $r \times 1$ column has $r \geq s$, then the ideal generated by its entries contains a power of $\mathfrak{m}$. This would make $\frac{R}{I}$ zero-dimensional. Hence $r < s$.

Choose a determinacy order k , and choose $N \geq k + 1$. Let $Z = \operatorname{Spec}\left(\frac{P}{\mathfrak{m}_0^N}\right)$. For each i , let $h_i^{<N} \in \mathfrak{m}_0$ be the polynomial truncation of h_i modulo $\mathfrak{m}^N$. Its class in $\frac{P}{\mathfrak{m}_0^N} = H^0(Z, \mathcal{O}_Z)$ is the Taylor data to be prescribed.

We construct polynomials $g_1, \dots, g_r$ successively, each satisfying $g_i \equiv h_i^{<N} \pmod{\mathfrak{m}_0^N}$, equivalently $g_i - h_i \in \mathfrak{m}^N R$ after completion. Put $X_0 = \mathbb{A}_K^s$, and after choosing the first i equations put $X_i = V(g_1, \dots, g_i)$. We require $X_i - \{0\}$ to be smooth of pure dimension $s - i$.

The assertion holds for X_0 . Assuming it for X_{i-1} , apply Lemma 2.5 to X_{i-1} , with the fixed fat point Z and the prescribed Taylor class of $h_i^{<N}$. Since Z is supported at the origin, the smooth open set in [35, Theorem 1.2] is precisely $X_{i-1} - \{0\}$. We obtain g_i such that $X_i - \{0\}$ is smooth of pure dimension $s - i$. Induction gives $X_r - \{0\}$ smooth of dimension $s - r > 0$.

Let $J = \langle g_1, \dots, g_r \rangle S_0$. Since J has r generators, every minimal prime has height at most r . A minimal prime cannot be $\mathfrak{m}_0$, because $\operatorname{ht} \mathfrak{m}_0 = s > r$. Therefore each irreducible component meets the punctured spectrum. On the punctured spectrum $\frac{S_0}{J}$ is smooth of pure dimension $s - r$, hence each minimal prime has height exactly r . Thus J has height r . Since S_0 is regular, hence Cohen–Macaulay, and J has height r , the r generators form a regular sequence [38, Lemmas 10.106.3 and 10.104.2, Tags 00NQ and 02JN].

Because a finite field is perfect, the Jacobian criterion applies [38, Lemma 10.137.15, Tag 00TE]. Let $\operatorname{JacMat}(g)$ be the $r \times s$ Jacobian matrix. Smoothness of $X_r - \{0\}$ implies that $J + I_r(\operatorname{JacMat}(g))$ is either the unit ideal or has radical $\mathfrak{m}_0$. Hence it contains a power of $\mathfrak{m}_0$. Since completion is faithfully flat, it preserves the regular sequence [38, Lemma 10.68.5, Tag 00LM]; the maximal-ideal-primary Jacobian condition is preserved as well. Consequently $\frac{R}{\langle g_1, \dots, g_r \rangle}$ is an ICIS in the sense of [13, Definition 3.1.27 and Theorem 3.1.28].

By construction, every $g_i - h_i$ lies in $\mathfrak{m}^N \subseteq \mathfrak{m}^{k+1}$. Finite contact k -determinacy therefore gives an isomorphism of complete local K -algebras $\frac{R}{I} \cong \frac{R}{\langle g_1, \dots, g_r \rangle}$. The complete-intersection property and regularity away from the closed point are intrinsic properties of this local algebra. Hence $\frac{R}{I}$ is an ICIS. $\square$

The converse is already part of the arbitrary-field theory summarized in [13, Theorem 3.1.28]. Combining that theorem with the known infinite-field case and Theorem 2.9 gives a field-independent statement. Thus no assumption on the cardinality of the ground field remains in the contact-determinacy criterion.

Corollary 2.10. *Let K be an arbitrary field and let $I \subseteq R$ be proper with $\dim(\frac{R}{I}) > 0$. Then I is finitely contact determined if and only if $\frac{R}{I}$ is an ICIS.*

We next consider matrix germs. The natural stratification for the left–right action is the stratification by rank. We first record its tangent spaces and then impose transversality to all rank strata simultaneously.

Let $Y = \text{Mat}_{m,n}(K)$ and set $d = mn$. For $0 \leq r \leq \min(m, n)$, let $\Sigma_r \subseteq Y$ be the smooth locally closed variety of matrices of rank exactly r .

Lemma 2.11. *For $C \in \Sigma_r$, one has $T_C \Sigma_r = M_m(K)C + CM_n(K)$.*

Proof. The rank- r stratum is the $\text{GL}_m(K) \times \text{GL}_n(K)$ -orbit of C . Differentiating the orbit map gives $M_m(K)C + CM_n(K) \subseteq T_C \Sigma_r$. Putting C into normal form with an $r \times r$ identity block shows that the left-hand side consists precisely of matrices whose lower-right $(m-r) \times (n-r)$ block vanishes. Its dimension is $r(m+n-r)$, which is the dimension of Σ_r . Therefore equality holds in every characteristic. $\square$

For a morphism $F : W \rightarrow V$ between smooth K -schemes and a smooth locally closed subscheme $S \subseteq V$, we say that F is *transverse to S at $w \in W$* with $F(w) \in S$ if $dF_w(T_w W) + T_{F(w)} S = T_{F(w)} V$; it is transverse to S on an open set if this holds at every point of the inverse image of S in that open set.

Theorem 2.12. *Let $V = \mathbb{A}_K^d = \coprod_{\alpha=1}^t S_\alpha$ be a finite decomposition into smooth locally closed K -subschemes. For $A \in \mathfrak{m}R^d$ and $N \geq 2$, there is $B \in A + \mathfrak{m}^N R^d$ such that:*

- (i) *B is the completion of an algebraic map $F : W \rightarrow V$ from an étale neighborhood $W \rightarrow \mathbb{A}_K^s$ of the origin;*
- (ii) *F is transverse to every S_α away from the distinguished point.*

Proof. Write the coordinates of V as $y_1, \dots, y_d$, put $E = \mathbb{A}_K^s \times V$, and let $z = (0, 0)$. Write $A = (A_1, \dots, A_d)$, and let $A_\beta^{<N}$ be the polynomial truncation of A_β modulo $\mathfrak{m}^N$.

We construct polynomial equations $h_1, \dots, h_d$ recursively so that h_β has the prescribed $(N-1)$ -jet $y_\beta - A_\beta^{<N}(x)$ at z . Put $\Gamma_i = V(h_1, \dots, h_i)$. The inductive requirement is that, away from z , Γ_i and every nonempty $\Gamma_i \cap (\mathbb{A}_K^s \times S_\alpha)$ are smooth of the expected dimensions. Assume $h_1, \dots, h_{i-1}$ have been chosen. We regard the relevant affine schemes as quasiprojective subschemes of one projective space and homogenize the polynomials. This allows us to use one ambient linear system throughout; no condition is imposed at infinity.

Suppose first that K is finite. By the inductive hypothesis, the quasiprojective schemes

$$X_0 = \Gamma_{i-1} \setminus \{z\}, \quad X_\alpha = (\Gamma_{i-1} \cap (\mathbb{A}_K^s \times S_\alpha)) \setminus \{z\}$$

are smooth whenever nonempty; zero-dimensional members are allowed. We use these finitely many schemes as the X_j in Lemma 2.6. At z we prescribe the unique coset modulo

the N -th power of the maximal ideal giving the required $(N - 1)$ -jet. This coset is open and closed in the completed local ring and hence has positive Haar measure and boundary of measure zero.

Now let $P \neq z$ be a closed point. For every positive-dimensional X_j containing P we allow precisely those first jets for which either $h_i(P) \neq 0$, or $h_i(P) = 0$ and

$$dh_i(P)|_{T_P X_j} \neq 0.$$

This is exactly the condition that the hypersurface section of X_j be smooth of dimension $\dim X_j - 1$ at P . If X_j is zero-dimensional, the expected dimension is -1 , and we require $h_i(P) \neq 0$. At points lying on none of the X_j there is no condition. For each fixed X_j containing P , the bad first jets form a proper linear subspace of the value-and-first-derivative space. Imposing the condition simultaneously for all X_j through P removes a finite union of such proper linear subspaces. In particular, the allowed set is a finite-jet cylinder of positive local measure—every jet with $h_i(P) \neq 0$ is allowed—and its boundary has measure zero. At a point of a stratum S_α , the tangent space of $\Gamma_{i-1} \cap (\mathbb{A}_K^s \times S_\alpha)$ is contained in $T_P \Gamma_{i-1}$. Hence, whenever both conditions occur, the imposed condition on the stratified intersection is compatible with the ambient smoothness condition.

All hypotheses of Lemma 2.6 are therefore satisfied, and an admissible h_i exists. This is the same closed-point-sieve mechanism used in [35, Theorem 3.3], now with the finite Taylor condition at z kept independent of the punctured smoothness conditions.

Now suppose that K is infinite. Let L_D be the affine space of polynomials of degree at most D with the prescribed jet at z . For $D \gg 0$, its translation space separates first jets at every geometric point of $E \setminus \{z\}$. Indeed, at a point $P \neq z$ one may choose a linear form vanishing at z and nonvanishing at P , raise it to the N -th power, and multiply by affine linear functions to obtain arbitrary value and tangent data at P . Let X be one of the finitely many smooth schemes occurring in the induction and put $e = \dim X$. For fixed $P \in X$, failure of a hypersurface section to have the expected smoothness is given by the $e + 1$ independent linear conditions $h_i(P) = 0$ and $dh_i(P)|_{T_P X} = 0$ when $e > 0$, and by the single condition $h_i(P) = 0$ when $e = 0$. Hence the corresponding incidence variety in $L_D \times X$ has dimension at most $\dim L_D - 1$. The Zariski closure of its projection in L_D therefore has dimension at most $\dim L_D - 1$, and is a proper closed subset. Removing the finitely many bad projections leaves a nonempty Zariski-open subset of L_D , and an infinite field has a K -rational point in this open set. Choosing that point completes the induction.

After d steps put $\Gamma = \Gamma_d$. At z we have $\left(\frac{\partial h_\alpha}{\partial y_\beta}\right)(z) = I_d$, so the projection $\Gamma \rightarrow \mathbb{A}_K^s$ is étale at z . The formal implicit-function theorem identifies the completed germ of Γ at z with the graph of a unique formal map $B(x)$, and the prescribed jets give $B - A \in \mathfrak{m}^N R^d$. After shrinking to an étale neighborhood W of the distinguished point, the y -coordinate map $F : W \rightarrow V$ is transverse to every S_α away from that point. $\square$

Theorem 2.13. *Let K be an arbitrary field, let $A \in \mathfrak{m}M_{m,n}(K[[x_1, \dots, x_s]])$, and let $N \geq 2$. There exists B with the same $(N-1)$ -jet as A such that the extended $\mathcal{G}_{lr}$ -tangent image of B has finite K -codimension.*

Proof. Apply Theorem 2.12 to the finite rank stratification $Y = \coprod_r \Sigma_r$. At every point away from the origin, transversality and Lemma 2.11 therefore give that $dB(T) + M_m B + B M_n = M_{m,n}$. Equivalently, the coherent homomorphism

$$M_m(\mathcal{O}_W) \oplus M_n(\mathcal{O}_W) \oplus T_W \longrightarrow M_{m,n}(\mathcal{O}_W), \quad (U, V, \xi) \longmapsto UB + BV + dB(\xi),$$

is surjective away from the distinguished point. Its cokernel is therefore supported at that point and has finite length. Completion identifies its image with the extended tangent image of the formal germ B . $\square$

The ordinary tangent image differs from the extended tangent image only by replacing the R -span of the finitely many derivative matrices by their $\mathfrak{m}$ -span. Their quotient is a quotient of a finite-dimensional K -space generated by the classes of the s derivative matrices. Therefore finite codimension of the ordinary and extended tangent images are equivalent.

Corollary 2.14. *For every $A \in \mathfrak{m}M_{m,n}$ and every N there exists $B \equiv A \pmod{\mathfrak{m}^N}$ that is finitely $\mathcal{G}_{lr}$ -determined.*

Proof. Theorem 2.13 gives B with finite tangent-image codimension. The arbitrary-field sufficient criterion [13, Theorem 3.1.12] then gives finite determinacy. $\square$

Theorem 2.15. *For every field K and every $A \in \mathfrak{m}M_{m,n}(K[[x_1, \dots, x_s]])$, finite $\mathcal{G}_{lr}$ -determinacy implies finite codimension of the tangent image. Together with the sufficient criterion of Greuel–Pham, this gives the equivalence predicted by Conjecture 3.1.16 for the full left–right group.*

Proof. Suppose A is k - $\mathcal{G}_{lr}$ -determined, and choose $N \geq k+1$. By Theorem 2.13, choose B with $B - A \in \mathfrak{m}^N M_{m,n}$ and with tangent image of finite codimension. Finite determinacy gives $A \sim_{\mathcal{G}_{lr}} B$. Finite codimension of the tangent image is invariant under $\mathcal{G}_{lr}$ -equivalence, by the chain rule and invertibility of the row, column, and coordinate transformations. Hence the tangent image of A has finite codimension. The converse is the arbitrary-characteristic sufficient criterion [13, Theorem 3.1.12]. $\square$

Combining the necessity proved above with the arbitrary-characteristic sufficient criterion gives a field-independent equivalence for the full left–right action. We record it explicitly because it will be used in the one-sided and determinantal arguments below:

$$A \text{ finitely } \mathcal{G}_{lr}\text{-determined} \iff \dim_K \left(\frac{M_{m,n}}{\tilde{T}_A^e(\mathcal{G}_{lr}A)} \right) < \infty.$$

This extends the previously known one-column result and the 2×2 infinite-field result; see [13, Theorem 3.1.23] and [34]. For the corresponding tangent-image formalism for right,

contact, and left–right equivalence, see also [13, §3.1].

The argument for matrices uses only the existence of a finite stratification whose tangent spaces are infinitesimal orbit directions. We record this observation in an abstract form before returning to the one-sided matrix actions. Let H be a smooth affine algebraic group acting on a finite-dimensional K -vector space V . For $v \in V$, write $\mathfrak{h} \cdot v = \text{ima}(d_e(H \rightarrow H \cdot v)) \subseteq T_v V$.

Definition 2.16. A finite smooth stratification $V = \coprod_{\alpha=1}^t S_\alpha$ is called an *infinitesimal-orbit stratification* if, after every field extension, at every point $v \in S_\alpha$ we have that $\mathfrak{h} \cdot v = T_v S_\alpha$.

The *current group* associated with the action $H \curvearrowright V$ is $\mathcal{G}_H = H(R) \rtimes \text{Aut}_K(R)$. Here an element of $H(R)$ acts pointwise on the target value of a formal map $A : (K^s, 0) \rightarrow V$, while $\text{Aut}_K(R)$ acts by change of source coordinates; the semidirect product records the resulting change of the R -valued point of H . Its extended infinitesimal current-orbit module is $\tilde{T}_A^e = R\langle \mathfrak{h} \cdot A \rangle + \sum_{\nu=1}^s R \frac{\partial A}{\partial x_\nu}$.

Theorem 2.17. Assume that V admits a finite infinitesimal-orbit stratification. Then, by Theorem 2.12, every finitely $\mathcal{G}_H$ -determined formal map A satisfies

$$\dim_K \left(\frac{V \otimes_K R}{\tilde{T}_A^e} \right) < \infty.$$

Whenever finite codimension of this infinitesimal module is known to be sufficient for finite determinacy for the action under consideration, it is therefore equivalent to finite determinacy.

Proof. Theorem 2.12 shows that every $\mathfrak{m}$ -adic cylinder around A contains a germ B transverse away from the origin to all strata. At a point with $B(x) \in S_\alpha$, we have that $dB_x(T_x) + T_{B(x)} S_\alpha = T_{B(x)} V$. By the infinitesimal-orbit hypothesis, we therefore get that $dB_x(T_x) + \mathfrak{h} \cdot B(x) = T_{B(x)} V$. Therefore the coherent cokernel of the infinitesimal current-orbit map is supported only at the origin, so its completion has finite length. Finite codimension of the infinitesimal module is invariant under the current action. Lemma 2.2 now transports the property from B to every finitely determined A . $\square$

Remark 2.18. For $V = M_{m,n}$ and $H = \text{GL}_m \times \text{GL}_n$, the rank strata form an infinitesimal-orbit stratification by Lemma 2.11. Thus Theorem 2.15 is the basic matrix instance of Theorem 2.17.

We next study the one-sided matrix actions. Once rank transversality is known, the remaining directions are measured by the kernel and image maps to Grassmannians.

For $1 \leq t \leq \min(m, n)$, let $I_t(A) \subseteq R$ denote the t -th *determinantal ideal*, generated by the $t \times t$ minors of A . If A is a formal matrix germ, we say that it is *transverse to the rank strata on the punctured formal spectrum* if for every nonmaximal prime $\mathfrak{p} \subseteq R$, after base change to the residue field $\kappa(\mathfrak{p})$, the induced map satisfies the tangent-space transversality

condition above at the corresponding point. Equivalently, the left–right infinitesimal orbit map is surjective after localization at every $\mathfrak{p} \neq \mathfrak{m}$.

For C in the rank- r stratum $\Sigma_r \subseteq M_{m,n}$ define $\kappa_r(C) = \ker C \in \mathrm{Gr}(n-r, n)$ and $\iota_r(C) = \mathrm{ima} C \in \mathrm{Gr}(r, m)$.

When a one-dimensional completed determinantal locus is reduced, by a *formal branch* we mean one of its irreducible components, equivalently one of its minimal primes. At the generic point $\mathfrak{q}$ of such a branch, the *continuous cotangent space* means the one-dimensional cotangent space obtained from continuous K -differentials on the completed local algebra and then tensoring with $\kappa(\mathfrak{q})$. In dimension one every punctured prime of a reduced local ring is minimal, so checking a differential on these branch generic points checks it on the whole punctured spectrum.

Lemma 2.19. *At $C \in \Sigma_r$, the kernel of $d\kappa_r$ is $M_m C$ and the kernel of $d\iota_r$ is $C M_n$. Hence the quotient of $T_C \Sigma_r$ by the left matrix orbit is $T_{\kappa_r(C)} \mathrm{Gr}(n-r, n)$, while the corresponding quotient by the right matrix orbit is $T_{\iota_r(C)} \mathrm{Gr}(r, m)$.*

Proof. The fibers of κ_r are the GL_m -orbits inside Σ_r , while the fibers of ι_r are the GL_n -orbits. Both maps are smooth homogeneous fibrations. Differentiating gives the statement. $\square$

Proposition 2.20. *Assume that A is transverse to all rank strata on the punctured spectrum. Then the extended $\mathcal{G}_l$ tangent image has finite codimension exactly when every induced kernel map $\kappa_r \circ A$ is submersive on $A^{-1}(\Sigma_r)$ away from the origin. Dually, the extended $\mathcal{G}_r$ tangent image has finite codimension exactly when all induced image maps $\iota_r \circ A$ are submersive there.*

Proof. Rank transversality supplies all normal directions to Σ_r . By Lemma 2.19, the directions missing from the left matrix orbit are exactly the tangent directions of the kernel Grassmannian, and they are supplied by the differential of $\kappa_r \circ A$. Thus localized surjectivity of the extended left tangent image is exactly submersivity of the kernel map. The right case is dual. Finite codimension is equivalent to localized surjectivity at every nonmaximal prime. $\square$

Corollary 2.21. *If A is finitely $\mathcal{G}_l$ -determined, then the only possible obstruction at this stage to finite codimension of its tangent image is a punctured critical locus of the kernel maps. Dually, for a finitely $\mathcal{G}_r$ -determined matrix the only possible obstruction at this stage is a punctured critical locus of the image maps.*

Proof. Finite one-sided determinacy implies finite $\mathcal{G}_{lr}$ -determinacy. Theorem 2.15 and Lemma 2.11 therefore give rank-stratum transversality away from the origin. Apply Proposition 2.20. $\square$

Proposition 2.22. *If $n \leq m$ and $I_n(A)$ is $\mathfrak{m}$ -primary, then A is finitely $\mathcal{G}_l$ -determined and its extended $\mathcal{G}_l$ tangent image has finite codimension. Dually, if $m \leq n$ and $I_m(A)$ is $\mathfrak{m}$ -primary, then A is finitely $\mathcal{G}_r$ -determined.*

Proof. At a punctured point, full column rank means that $C : K^n \rightarrow K^m$ is injective. Every $B : K^n \rightarrow K^m$ can then be written $B = UC$ for some $U \in M_m(K)$, so the left matrix directions already fill the target. Therefore the tangent cokernel has finite length, and the sufficient tangent-image criterion [13, Theorem 3.1.12] gives finite determinacy. The right case is the transpose. $\square$

Theorem 2.23. *If $n \leq m$ and $s \leq m - n + 1$, then finite $\mathcal{G}_l$ -determinacy, $\mathfrak{m}$ -primaryness of $I_n(A)$, and finite codimension of the extended $\mathcal{G}_l$ tangent image are equivalent. Dually, if $m \leq n$ and $s \leq n - m + 1$, the analogous three conditions for $\mathcal{G}_r$ and $I_m(A)$ are equivalent.*

Proof. Only necessity of the determinantal condition remains. Finite one-sided determinacy implies finite $\mathcal{G}_{lr}$ -determinacy. Theorem 2.37 gives $\text{ht } I_n(A) = \min\{s, m - n + 1\} = s$ in the left case, hence $I_n(A)$ is $\mathfrak{m}$ -primary. The right case is dual. The remaining implications are Proposition 2.22 and the sufficient criterion [13, Theorem 3.1.12]. $\square$

Lemma 2.24. (i) *If A is k - $\mathcal{G}_l$ -determined and $V - I_n \in \mathfrak{m}^k M_n(R)$, then $AV \sim_{\mathcal{G}_l} A$, and on every rank- r locus the kernel maps differ by a source automorphism.*

(ii) *Dually, if A is k - $\mathcal{G}_r$ -determined and $U - I_m \in \mathfrak{m}^k M_m(R)$, then $UA \sim_{\mathcal{G}_r} A$, and on every rank- r locus the image maps differ by a source automorphism.*

Proof. Since $A \in \mathfrak{m} M_{m,n}$, the condition $V - I_n \in \mathfrak{m}^k M_n(R)$ gives $AV - A \in \mathfrak{m}^{k+1} M_{m,n}$. Finite $\mathcal{G}_l$ -determinacy therefore gives $AV = U\varphi(A)$ for some $U \in \text{GL}_m(R)$ and $\varphi \in \text{Aut}_K(R)$. Right multiplication by V and left multiplication by U do not change matrix rank, and $I_t(AV) = I_t(A)$ while $I_t(U\varphi(A)) = \varphi(I_t(A))$. Hence φ preserves every determinantal rank locus. On the rank- r locus, left multiplication does not change the kernel and right multiplication sends the kernel by V^{-1} . Thus the two kernel maps differ only by the source automorphism induced by φ . The proof for $\mathcal{G}_r$ is obtained by transposing the argument: left gauges act on image subspaces, whereas right multiplication does not change them. $\square$

In positive characteristic, Frobenius gives sufficiently deep gauges whose differentials vanish. We first record the only form of this observation that will be needed below. For $N \geq 1$, call $V \in \text{GL}_n(R)$ *N -deep* if $V - I_n \in \mathfrak{m}^N M_n(R)$. For a polynomial local gauge over S_0 , we use the same terminology with $\mathfrak{m}_0 S_0$ in place of $\mathfrak{m}$.

Lemma 2.25. *Assume that $\text{char } K = p > 0$ and let $N \geq 1$. Let $\mathfrak{q} \neq \mathfrak{m}$ be a prime of R , and put $L = \text{Frac}\left(\frac{R}{\mathfrak{q}}\right)$. Then the values at $\mathfrak{q}$ of N -deep polynomial local gauges $V \in \text{GL}_n(S_0)$ satisfying $dV = 0$ are Zariski dense in $\text{GL}_n(L)$.*

Proof. Put $\eta = \mathfrak{q} \cap S_0$. Since $\mathfrak{q} \neq \mathfrak{m}$, we have that $\eta \neq \mathfrak{m}_0$. Therefore we may choose $a \in \mathfrak{m}_0^N \setminus \eta$. For $B = [b_{ij}] \in M_n(S_0)$, write $B^{[p]} = (b_{ij}^p)$, and set $V_B = I_n + a^p B^{[p]}$. Since $a \in \mathfrak{m}_0$, we get that $\det(V_B) \equiv 1 \pmod{\mathfrak{m}_0}$, so $V_B \in \text{GL}_n(S_0)$. Moreover,

$$V_B - I_n \in \mathfrak{m}_0^{Np} M_n(S_0) \subseteq \mathfrak{m}_0^N M_n(S_0),$$

and, since all entries added to the identity are p -th powers, we have that $dV_B = 0$. The domain S_0/η is infinite. Indeed, this is immediate when K is infinite, while when K is finite it follows from $\eta \neq \mathfrak{m}_0$, because then S_0/η has positive dimension. Frobenius is

injective on this domain, and therefore its set of p -th powers is infinite. Consequently its image in L is Zariski dense in $\mathbb{A}_L^1$. It follows entrywise that the values $I_n + \bar{a}^p B(\mathfrak{q})^{[p]}$ are Zariski dense in $M_n(L)$. Intersecting with the nonempty open subset $\mathrm{GL}_n(L)$ therefore proves the assertion. $\square$

For a morphism $F : X \rightarrow Y$ between smooth schemes, we write $\mathrm{Crit}(F)$ for the degeneracy subscheme on which dF is not surjective. Thus, when $\dim X = d$ and $\dim Y = t \leq d$, the critical scheme is defined by the maximal minors of a local matrix for $dF : T_X \rightarrow F^*T_Y$.

The following general-position lemma replaces the generic-smoothness step. Its point is that we control the dimension of the critical locus in the *source*; no assertion concerning an open subset of the target is required.

Lemma 2.26. *Let K be infinite, let $U \subseteq \mathbb{A}_K^s$ be an open neighborhood of the origin, and let $X \subseteq U \setminus \{0\}$ be a smooth locally closed K -subscheme of pure dimension d . Let $F : X \rightarrow \mathrm{Gr}(a, n)$ where $t = a(n - a)$ and assume $1 \leq t \leq d$. For every $N \geq 1$, there exists a polynomial matrix $V = I_n + H$ with $H \in M_n(\langle x_1, \dots, x_s \rangle^N K[x_1, \dots, x_s])$ such that on $X_V = X \cap D(\det V)$ we have that $\dim \mathrm{Crit}(V^{-1} \cdot F) \leq t - 1$. In particular, V defines an N -deep element of $\mathrm{GL}_n(S_0)$. If $t = 0$, the critical locus is empty.*

Proof. Fix $D \geq N + 1$, and let $E_D = \langle x_1, \dots, x_s \rangle^N \cap K[x_1, \dots, x_s]_{\leq D}$ and $\mathcal{P}_D = M_n(E_D)$. For every geometric point $x \neq 0$, evaluation of first jets gives a surjection $E_D \otimes_K \kappa(x) \rightarrow \mathcal{O}_{\mathbb{A}_{\kappa(x)}^s, x}^s / \mathfrak{m}_x^2$. Indeed, choose $x_\nu(x) \neq 0$. Multiplication by x_ν^N is invertible on the first-jet algebra, and affine-linear polynomials supply all first jets.

Consider the open subset $\Omega = \{(x, H) \in X \times \mathcal{P}_D : \det(I_n + H(x)) \neq 0\}$. On Ω , the expression $(I_n + H)^{-1} \cdot F$ is defined. Fix a geometric point x and an invertible value $C = I_n + H(x)$. Keeping this value fixed, the first derivatives of H can vary arbitrarily. Since the orbit map $\mathrm{GL}_n \rightarrow \mathrm{Gr}(a, n)$ is smooth, these variations induce arbitrary variations of the differential in $\mathrm{Hom}_{\kappa(x)}(T_x X, T_{C^{-1} \cdot F(x)} \mathrm{Gr}(a, n))$. The locus of maps from a d -dimensional space to a t -dimensional space that are not surjective has codimension $d - t + 1$.

Consequently, the critical incidence locus $\mathcal{Z} \subseteq \Omega$ satisfies

$$\dim \mathcal{Z} \leq \dim \mathcal{P}_D + d - (d - t + 1) = \dim \mathcal{P}_D + t - 1.$$

The generic fiber-dimension theorem, applied to the finitely many irreducible components of $\mathcal{Z}$, gives a nonempty open subset $\mathcal{U} \subseteq \mathcal{P}_D$ over which all fibers have dimension at most $t - 1$. Since K is infinite, choose $H \in \mathcal{U}(K)$. Finally, $V(0) = I_n$, so $\det V$ is a unit in S_0 . $\square$

We can now prove the one-sided criterion over infinite fields. The argument uses two different kinds of deep gauges. Lemma 2.26 produces a gauge for which the critical locus has dimension strictly smaller than the Grassmannian. Lemma 2.25 then shows that a nonempty critical locus would nevertheless have to move densely in the Grassmannian. These two conclusions are incompatible.

Theorem 2.27. *Let $A \in \mathfrak{m}M_{m,n}$. If K is infinite, then:*

- (i) *A is finitely $\mathcal{G}_l$ -determined if and only if $\dim_K \left(\frac{M_{m,n}}{\bar{T}_A^e(\mathcal{G}_l A)} \right) < \infty$.*
- (ii) *A is finitely $\mathcal{G}_r$ -determined if and only if $\dim_K \left(\frac{M_{m,n}}{\bar{T}_A^e(\mathcal{G}_r A)} \right) < \infty$.*

The same equivalences hold for the ordinary tangent images. Over an arbitrary field, including a finite field, the corresponding equivalences hold in the dimension ranges of Theorem 2.23.

Proof. Finite codimension of the tangent image is sufficient for finite determinacy over an arbitrary field by the Greuel–Pham finite-determinacy criterion [13, Theorem 3.1.12]. Over a finite field, the assertion in the dimension ranges stated above is precisely Theorem 2.23, together with the equivalence between ordinary and extended tangent-image codimension. It therefore remains to prove necessity when K is infinite. In characteristic zero, necessity follows from separability of the finite-jet orbit maps; see [13, Theorem 3.1.21] and [16, Theorem 4.3]. We may therefore assume that $\text{char } K = p > 0$. We prove the assertion for $\mathcal{G}_l$. The $\mathcal{G}_r$ case follows by transposition.

Assume that A is finitely $\mathcal{G}_l$ -determined, and let k be a determinacy order. Replacing A by a polynomial truncation in its $\mathcal{G}_l$ -orbit, we may suppose that the entries of A belong to S_0 . We use the same symbol A for this algebraic representative and for its completion. Finite $\mathcal{G}_l$ -determinacy implies finite $\mathcal{G}_{lr}$ -determinacy. Therefore Theorem 2.15, together with Corollary 2.36, shows that A is transverse to every constant-rank stratum on the punctured formal spectrum.

Fix a rank r . If the rank- r locus has no punctured point, there is nothing to prove. We therefore assume that it does. Put $c = (m-r)(n-r)$ and $d = s - c$. By Theorem 2.37, the rank-at-most- r locus has the expected codimension c , and its punctured rank- r stratum is smooth of pure dimension d . Let $F = \kappa_r \circ A$ be the kernel map on the rank- r stratum, and put $t = \dim \text{Gr}(n-r, n) = r(n-r)$. If $t = 0$, the Grassmannian is a point, and therefore there is no kernel-map obstruction. Thus assume that $t > 0$.

We carry out all incidence and image-dimension arguments on algebraic representatives in an open neighborhood $U \subseteq \mathbb{A}_K^s$ of the origin, before localizing. After shrinking U , the punctured rank- r locus is smooth of pure dimension $d = s - (m-r)(n-r)$. When passing to the local ring and its completion, we use the reduced closures of the relevant locally closed loci. Their completed germs are identified by the formal source automorphisms arising from finite determinacy. Completion preserves the dimension of these closed local germs. Components of an algebraic closure that do not contain the origin are removed by shrinking U . Thus the resulting dimension bounds also bound the algebraic image closures used below.

We first show that $t \leq d$. Let $X_{r,1}, \dots, X_{r,a}$ be the finitely many algebraic irreducible components of the punctured rank- r locus, and let $Y_j = \overline{F(X_{r,j})} \subseteq \text{Gr}(n-r, n)$ be their

image closures. Since $\dim X_{r,j} = d$, we have that $\dim Y_j \leq d$ for every j . Choose a generic prime $\mathfrak{q}$ of a formal irreducible component of the completed rank- r locus, and put $L = \text{Frac}\left(\frac{R}{\mathfrak{q}}\right)$.

Let V be a sufficiently deep polynomial gauge with $dV = 0$. Lemma 2.24 gives a formal source automorphism φ_V preserving every determinantal rank locus such that, on the completed rank- r locus we have that $V^{-1} \cdot F = F \circ \varphi_V$. The automorphism φ_V permutes the finitely many formal irreducible components. Every such formal component lies over one of the algebraic components $X_{r,j}$. Therefore $V(\mathfrak{q})^{-1} \cdot F(\mathfrak{q}) \in \bigcup_{j=1}^a (Y_j)_L$. By Lemma 2.25, the possible values $V(\mathfrak{q})$ are Zariski dense in $\text{GL}_n(L)$. Since GL_n acts transitively on $\text{Gr}(n-r, n)$, the corresponding points $V(\mathfrak{q})^{-1} \cdot F(\mathfrak{q})$ are Zariski dense in the Grassmannian after base change to L . Consequently

$$\text{Gr}(n-r, n)_L \subseteq \bigcup_{j=1}^a (Y_j)_L.$$

The Grassmannian is irreducible. Hence one of the Y_j is the whole Grassmannian. Therefore $t \leq d$, as claimed.

We may now apply Lemma 2.26 to the punctured algebraic rank- r locus. Taking the depth at least k , we obtain some $V \in \text{GL}_n(S_0)$ such that $V - I_n \in \mathfrak{m}_0^k M_n(S_0)$ and $\dim \text{Crit}(V^{-1} \cdot F) \leq t - 1$. Since $A \in \mathfrak{m} M_{m,n}$, we have that $AV - A \in \mathfrak{m}^{k+1} M_{m,n}$. Finite $\mathcal{G}_l$ - k -determinacy therefore gives $AV \sim_{\mathcal{G}_l} A$. By Lemma 2.24, the corresponding completed kernel maps differ by a formal source automorphism. The chain rule therefore identifies their completed critical schemes. Completion preserves the dimension of the corresponding local schemes, and hence the critical locus of the original completed kernel map also satisfies $\dim \text{Crit}(F) \leq t - 1$.

We claim that this critical locus is in fact empty on the punctured spectrum. Suppose otherwise. Let $Z_1, \dots, Z_b$ be the algebraic irreducible components of the critical locus of F on the punctured rank- r stratum. The preceding bound gives $\dim Z_i \leq t - 1$ for every i . Put $W_i = \overline{F(Z_i)} \subseteq \text{Gr}(n-r, n)$. Thus $\dim W_i \leq t - 1$.

Choose a generic prime $\mathfrak{q}$ of a formal irreducible component of the completed critical locus. Let V now range over sufficiently deep polynomial gauges satisfying $dV = 0$. Therefore, the differential of $V^{-1} \cdot F$ is obtained from dF by composition with an invertible linear map on the target tangent space. Therefore the critical scheme of $V^{-1} \cdot F$ is exactly the critical scheme of F .

On the other hand, Lemma 2.24 gives $V^{-1} \cdot F = F \circ \varphi_V$. Thus φ_V preserves the completed critical locus and permutes its finitely many formal irreducible components. Therefore, for $L = \text{Frac}\left(\frac{R}{\mathfrak{q}}\right)$, we get that

$$V(\mathfrak{q})^{-1} \cdot F(\mathfrak{q}) \in \bigcup_{i=1}^b (W_i)_L.$$

Lemma 2.25 again shows that the values $V(\mathfrak{q})$ are Zariski dense in $\mathrm{GL}_n(L)$. Hence the left-hand side is Zariski dense in $\mathrm{Gr}(n-r, n)_L$. We would therefore obtain $\mathrm{Gr}(n-r, n)_L \subseteq \bigcup_{i=1}^b (W_i)_L$. This is impossible since the left-hand side is irreducible of dimension t , whereas every W_i has dimension at most $t-1$. Thus we can conclude that $\mathrm{Crit}(F) = \emptyset$ on the punctured rank- r locus. Therefore, we can conclude that every kernel map of the form $\kappa_r \circ A$ is submersive on the corresponding punctured constant-rank stratum. Proposition 2.20 therefore implies that the extended $\mathcal{G}_l$ tangent image has finite K -codimension. Finally, the ordinary tangent image differs from the extended tangent image only by replacing the R -span of the finitely many derivative matrices by their $\mathfrak{m}$ -span. The quotient between these two derivative contributions is finite-dimensional over K . Thus finite codimension of the ordinary and extended tangent images are equivalent. $\square$

Remark 2.28. Over an infinite field, Theorem 2.15, Theorem 2.32, and Theorem 2.27 give necessity and sufficiency of the tangent-image criterion for all four actions, namely $\mathcal{R}, \mathcal{G}_l, \mathcal{G}_r, \mathcal{G}_{lr}$. Over a finite field, the full left-right and pure right criteria remain valid over arbitrary fields, while the one-sided equivalence is established in the dimension ranges of Theorem 2.23. The unrestricted finite-field one-sided case would require an additional finite-field general-position theorem replacing Lemma 2.26.

Remark 2.29. Finite codimension of the tangent image can be checked by the standard-basis methods developed by Greuel-Pham [18]. Consequently, over an infinite field the preceding theorems give a decision criterion for finite determinacy for all four matrix actions whenever the ground field is given in an effective form. Over finite fields, the same conclusion holds for the full left-right and pure right actions and, for the one-sided actions, in the ranges of Theorem 2.23.

We next consider pure right equivalence for vector-valued germs. When the target has dimension greater than one, finite right determinacy is much more rigid. For a vector-valued germ $F \in \mathfrak{m}R^q$, *pure right equivalence* means equivalence under source coordinate changes only: F and G are right equivalent if $G = F \circ \varphi$ for some $\varphi \in \mathrm{Aut}_K(R)$; no target coordinate change is allowed. Write $F = (F_1, \dots, F_q)$ and let $\mathrm{JacMat}(F) = \left(\frac{\partial F_i}{\partial x_j} \right)_{i,j}$ be its $q \times s$ Jacobian matrix.

Lemma 2.30. *Let $T = K[[y_1, \dots, y_q]]$ with maximal ideal $\mathfrak{m}_T$. Suppose a proper ideal $D \subseteq T$ is invariant under every sufficiently deep triangular automorphism and contains a nonzero polynomial. Then $\sqrt{D} = \mathfrak{m}_T$.*

Proof. Choose $0 \neq P \in D \cap K[y_1, \dots, y_q]$. Fix j . For $i \neq j$ choose integers M_i larger than the depth from which triangular automorphisms preserve D and, successively, so large that the map $\alpha \mapsto \alpha_j + \sum_{i \neq j} M_i \alpha_i$ is injective on the finite support of P . Apply the triangular automorphism σ_j given by $y_i \mapsto y_i + y_j^{M_i}$ for $i \neq j$ and $y_j \mapsto y_j$. In $\sigma_j(P)$, select from each substituted factor the pure y_j -term. Injectivity of the displayed weight on the support implies that one original monomial contributes a unique largest pure power of y_j , with nonzero coefficient. Fix once and for all a degree-compatible monomial order on $K[y_1, \dots, y_q]$. Because every $M_i > 1$, within the expansion of any fixed monomial of P the term obtained by choosing the pure $y_j^{M_i}$ contribution from each substituted factor has

strictly largest total degree. Choosing the M_i successively and sufficiently large makes these maximal total degrees distinct for the finitely many monomials in the support of P . Thus $\sigma_j(P)$ has a unique highest-total-degree term, and that term is a positive power of y_j ; in particular it is the leading monomial for the fixed degree-compatible order.

Put $J = D \cap K[y_1, \dots, y_q]$. The initial ideal of J contains a power of every variable, so $K[y]/J$ is zero-dimensional. Since $J \subseteq \langle y \rangle$, its localization at $\langle y \rangle$ is a proper zero-dimensional local ring. Therefore, for some b , we have that $\langle y \rangle^b K[y]_{\langle y \rangle} \subseteq JK[y]_{\langle y \rangle}$. Extending to $K[[y]]$ gives $\mathfrak{m}_T^b \subseteq D$, and therefore $\sqrt{D} = \mathfrak{m}_T$. $\square$

Lemma 2.31. *Assume $2 \leq q \leq s$ and let $F \in \mathfrak{m}K[x]^q$ satisfy $\text{rank}(\text{JacMat}(F)(0)) < q$. For every N there is $H \in \mathfrak{m}^N K[x]^q$ such that, for $G = F + H$, the ideal $I_q(\text{JacMat}(G))$ has height $s - q + 1$ after localization and completion at the origin. Moreover the critical-value ideal*

$$\Delta_G = \ker \left(K[[y_1, \dots, y_q]] \longrightarrow \frac{R}{I_q(\text{JacMat}(G))}, \quad y_i \longmapsto G_i \right)$$

contains a nonzero polynomial.

Proof. First replace N by a larger integer if necessary. Choose pairwise distinct integers $a_1, \dots, a_q \geq N$, none divisible by $\text{char } K$, and larger than the degrees occurring in F , and set $H_i = \sum_{j=1}^s x_j^{a_i}$. For a q -element subset $J = \{j_1, \dots, j_q\}$ of the source variables, the part of highest total degree in the corresponding $q \times q$ Jacobian minor is $\det[a_i x_{j_\ell}^{a_i-1}]_{1 \leq i, \ell \leq q}$. This generalized alternant is nonzero: after expansion, distinct permutations give distinct exponent vectors because the a_i are distinct, and every coefficient $\prod_i a_i$ is nonzero in K . Choose a degree-compatible monomial order. For every q -element subset J , the leading monomial of the corresponding Jacobian minor is therefore a monomial of the highest-total-degree alternant. Its support is exactly J , and each variable indexed by J has positive exponent. Its leading monomial has a strictly positive exponent on every variable indexed by J and no other variables. Therefore a positive power of $\prod_{j \in J} x_j$ belongs to the initial ideal. Hence the radical of the initial ideal of $I_q(\text{JacMat}(G))$ contains the squarefree monomial ideal generated by $\prod_{j \in J} x_j$ over all q -element subsets J . That squarefree ideal has height $s - q + 1$. A Gröbner degeneration preserves Krull dimension [9, Chapter 15], so $\text{ht } I_q(\text{JacMat}(G)) = \text{ht in}(I_q(\text{JacMat}(G))) \geq s - q + 1$; the standard determinantal height bound [7, Chapter 2] gives the reverse inequality. Thus equality holds in $K[x_1, \dots, x_s]$. Since $\text{rank}(\text{JacMat}(G)(0)) < q$, every maximal minor vanishes at the origin, so $I_q(\text{JacMat}(G)) \subseteq \langle x_1, \dots, x_s \rangle$. Every prime contained in this maximal ideal and minimal over $I_q(\text{JacMat}(G))$ has height at least $s - q + 1$, while the determinantal height bound remains valid after localization. Hence the localized ideal has height exactly $s - q + 1$. Completion preserves the dimension of the quotient [38, Lemma 15.44.1, Tag 07NV], and therefore preserves the height in this regular local ring. This proves the stated local and completed assertion.

Every minimal component of the critical locus consequently has dimension at most $q - 1$. The restrictions of the q functions $G_1, \dots, G_q$ to such a component are algebraically dependent, so for each minimal component there is a nonzero polynomial $P(y_1, \dots, y_q)$ vanishing

on its image. Multiplying these relations over the finitely many minimal components gives a nonzero polynomial in the radical of the kernel of $K[y_1, \dots, y_q] \longrightarrow \frac{K[x_1, \dots, x_s]}{I_q(\text{JacMat}(G))}$, $y_i \mapsto G_i$. Taking a suitable power produces a nonzero polynomial in the kernel itself. This is the asserted critical-value relation. $\square$

Theorem 2.32. *For every field K and every $F \in \mathfrak{m}R^q$,*

$$F \text{ is finitely } \mathcal{R}\text{-determined} \iff \dim_K \left(\frac{R^q}{\text{ima}(\text{JacMat}(F))} \right) < \infty.$$

If $q > 1$, these conditions are equivalent to $\text{rank}(\text{JacMat}(F)(0)) = q$. In that case $q \leq s$, the germ F is right equivalent to the coordinate projection $(x_1, \dots, x_q)$, and F is already 1- $\mathcal{R}$ -determined.

Proof. For $q = 1$, the assertion is the arbitrary-characteristic finite-Milnor-number criterion [6, Theorem 2.1]. Assume $q > 1$. Proposition 2.4 gives $q \leq s$. After replacing F by a polynomial truncation in its right orbit, define its *relation ideal* and *critical-value ideal* by $E_F = \ker(K[[y_1, \dots, y_q]] \rightarrow R, y_i \mapsto F_i)$ and $\Delta_F = \ker(K[[y_1, \dots, y_q]] \rightarrow \frac{R}{I_q(\text{JacMat}(F))}, y_i \mapsto F_i)$, respectively. Let k be a right-determinacy order for F . If a target automorphism ψ satisfies $\psi(y) - y \in \mathfrak{m}_T^{k+1}T^q$, then $F_i \in \mathfrak{m}$ gives $\psi \circ F - F \in \mathfrak{m}^{k+1}R^q$. Hence $\psi \circ F$ is right equivalent to F , so there is a source automorphism φ with $\psi \circ F = F \circ \varphi$. For such ψ , the chain rule gives $\text{JacMat}(\psi \circ F) = D\psi(F) \text{JacMat}(F)$ with $D\psi(F)$ invertible. Thus the maximal-minor ideal is unchanged up to the induced source automorphism, and both E_F and Δ_F are invariant under these deep target automorphisms.

We first record the following consequence of right determinacy, which we use in the proof: the components of F minimally generate the ideal $\langle F_1, \dots, F_q \rangle$. Indeed, suppose after reordering that $F_q \in I_0 = \langle F_1, \dots, F_{q-1} \rangle$. Since $q \leq s$, the ideal I_0 has at most $s - 1$ generators and therefore is not $\mathfrak{m}$ -primary. Thus, for every N , one can choose $h \in \mathfrak{m}^N \setminus I_0$. Take N larger than a right-determinacy order and set $G = (F_1, \dots, F_{q-1}, F_q + h)$. Then $G = \varphi(F)$ for some source automorphism φ . For $i < q$ this identity gives $\varphi(F_i) = F_i$, whereas applying φ to a relation $F_q = \sum_{i < q} a_i F_i$ gives $F_q + h = \sum_{i < q} \varphi(a_i) F_i$, so $h \in I_0$, a contradiction.

Suppose $\text{rank}(\text{JacMat}(F)(0)) < q$. Apply Lemma 2.31 beyond the determinacy order. The resulting right-equivalent G has a critical-value ideal containing a nonzero polynomial, so Lemma 2.30 gives $\sqrt{\Delta_G} = \mathfrak{m}_T$. Thus every component G_i belongs to the radical of $I_q(\text{JacMat}(G))$.

Right determinacy implies finite contact determinacy of the generator column. If $q < s$, Corollary 2.10 shows that $\langle G_1, \dots, G_q \rangle$ defines an ICIS. Therefore $\langle G_1, \dots, G_q \rangle + I_q(\text{JacMat}(G))$ is $\mathfrak{m}$ -primary. Since every G_i lies in $\sqrt{I_q(\text{JacMat}(G))}$, these two ideals have the same radical, and $I_q(\text{JacMat}(G))$ would itself be $\mathfrak{m}$ -primary. If $q = s$ and $\langle G_1, \dots, G_q \rangle$ were not $\mathfrak{m}$ -primary, then it would have positive-dimensional quotient; Corollary 2.10 would make its $q = s$ minimal generators a regular sequence, forcing a zero-dimensional

quotient, a contradiction. Thus $\langle G_1, \dots, G_q \rangle$ is $\mathfrak{m}$ -primary. The same radical containment therefore forces $I_q(\text{JacMat}(G))$ to be $\mathfrak{m}$ -primary. Both alternatives are impossible because a proper maximal-minor ideal of a $q \times s$ matrix has height at most $s - q + 1 < s$. Hence $\text{rank}(\text{JacMat}(F)(0)) = q$.

A full-rank Jacobian at the origin makes F a formal submersion, so a source coordinate change sends it to $(x_1, \dots, x_q)$. Any germ with the same 1-jet as F again has Jacobian rank q at the origin and is therefore right equivalent to the same coordinate projection. Hence F is 1- $\mathcal{R}$ -determined. Conversely, if $\frac{R^q}{\text{ima}(\text{JacMat}(F))}$ has finite length, then the Jacobian map is surjective after localization at every nonmaximal prime. Thus no nonmaximal prime contains the ideal $I_q(\text{JacMat}(F))$ of maximal minors, and hence $I_q(\text{JacMat}(F))$ is either $\mathfrak{m}$ -primary or the unit ideal. Tensoring with $\text{Frac}(R)$ also shows that $\text{JacMat}(F)$ has row rank q , so necessarily $q \leq s$. The determinantal height bound [7, Chapter 2] $\text{ht } I_q(\text{JacMat}(F)) \leq s - q + 1 < s$ excludes the proper $\mathfrak{m}$ -primary case when $q > 1$, so $I_q(\text{JacMat}(F)) = R$. $\square$

Remark 2.33. The rigidity in Theorem 2.32 is specific to pure right equivalence. When $q > 1$, a finitely right-determined map germ cannot have a singular differential at the origin: it is already a formal submersion and hence a coordinate projection. For example,

$$(x_1^2, x_2, \dots, x_q) : (K^s, 0) \longrightarrow (K^q, 0)$$

is never finitely pure-right determined for $q > 1$, in any characteristic. This should be contrasted with left-right and contact equivalence, where non-submersive finitely determined germs are the principal objects of singularity theory.

Corollary 2.34. *Let (Y, y) be smooth with $y \in Y(K)$ and $\dim_y Y = q > 1$, and let $F : (\text{Spf } R, \mathfrak{m}) \rightarrow (\hat{Y}_y, y)$ be a formal map germ. Then F is finitely right determined if and only if $dF_0 : T_0 \mathbb{A}_K^s \rightarrow T_y Y$ is surjective. In this case F is 1-right-determined. These statements do not depend on the choice of formal parameters at y .*

Proof. Since Y is smooth at the K -rational point y , its completed local ring is (noncanonically) isomorphic to $K[[y_1, \dots, y_q]]$. Under such an isomorphism the assertion is exactly Theorem 2.32. Replacing one regular system of parameters by another composes the coordinate representative with a target formal automorphism; this preserves surjectivity of dF_0 and the property that all germs with the same 1-jet lie in one source-coordinate orbit. $\square$

Corollary 2.35. *Let $A \in \mathfrak{m}M_{m,n}$ over an arbitrary field. Then A is finitely $\mathcal{R}$ -determined if and only if its ordinary, equivalently its extended, $\mathcal{R}$ -tangent image has finite K -codimension. If $mn > 1$, these conditions are equivalent to surjectivity at the origin of the differential of the entry map $A : (K^s, 0) \longrightarrow (K^{mn}, 0)$, and in this case A is already 1- $\mathcal{R}$ -determined. Consequently, Theorems 2.15 and 2.32 settle the full left-right and pure right cases of GLS Conjecture 3.1.16 over every field. Together with Theorem 2.27, they settle all four actions over every infinite field. Over an arbitrary field, the one-sided cases are additionally settled in the dimension ranges of Theorem 2.23.*

Proof. Vectorize the entries of A and apply Theorem 2.32 with $q = mn$. The extended $\mathcal{R}$ -tangent image is the R -span of the derivative matrices, whereas the ordinary $\mathcal{R}$ -tangent image is their $\mathfrak{m}$ -span. The quotient of the former derivative contribution by the latter is generated over K by the finitely many derivative matrices. Therefore finite codimension of the ordinary tangent image is equivalent to finite codimension of the extended tangent image.

If $mn > 1$, Theorem 2.32 shows that finite right determinacy is equivalent to having $\text{rank}(\text{JacMat}(A)(0)) = mn$. This is exactly surjectivity at the origin of the differential of the entry map $A : (K^s, 0) \rightarrow (K^{mn}, 0)$. The same theorem then shows that A is 1- $\mathcal{R}$ -determined. For $mn = 1$, the assertion is the scalar finite-right-determinacy criterion. $\square$

Finally, we translate punctured rank transversality into statements about determinantal ideals. The homological study of determinantal ideals begins with the Eagon–Northcott complex [8]; the expected-height perfection and Cohen–Macaulayness results were subsequently developed in the generic-perfection work of Hochster–Eagon [22]. We use the modern treatment in [7]. For $1 \leq t \leq \min(m, n)$ put $c_t = (m - t + 1)(n - t + 1)$.

Corollary 2.36. *A matrix germ is finitely $\mathcal{G}_{lr}$ -determined exactly when, on the punctured formal spectrum, it is transverse to every constant-rank stratum.*

Proof. By Lemma 2.11, punctured transversality is equivalent to localized surjectivity of the extended left–right tangent image. For a finitely generated cokernel over R , this is equivalent to finite length. Apply Theorem 2.15. $\square$

Theorem 2.37. *Let $G \in \{\mathcal{R}, \mathcal{G}_l, \mathcal{G}_r, \mathcal{G}_{lr}\}$ and let A be finitely G -determined. Then*

$$\text{ht } I_t(A) = \min\{s, c_t\} \quad (1 \leq t \leq \min(m, n)).$$

Moreover $\frac{R}{I_t(A)}$ is Cohen–Macaulay; if $c_t < s$ it is reduced; and if $c_t \leq s - 2$ it is a normal domain.

Proof. Finite G -determinacy implies finite $\mathcal{G}_{lr}$ -determinacy. At every nonmaximal point lying over $I_t(A)$, Corollary 2.36 gives the expected inverse-image codimension. Combined with the standard determinantal height bound [7, Chapter 2], this proves the height formula. If $c_t \leq s$, the equality $\text{ht } I_t(A) = c_t$ is the expected-height case, so determinantal perfection gives Cohen–Macaulayness [7, Chapter 2]. If $c_t > s$, then $\text{ht } I_t(A) = s$ and $\frac{R}{I_t(A)}$ is zero-dimensional, hence Cohen–Macaulay.

If $c_t < s$, every minimal prime lies in the punctured spectrum and has generic matrix rank $t - 1$. Transversality makes the quotient smooth at every generic point. It is therefore generically reduced. Since a Cohen–Macaulay ring has no embedded associated primes [38, Lemma 10.103.7, Tag 0BUS], it is reduced.

Assume $c_t \leq s - 2$. Then

$$\dim \left(\frac{R}{I_t(A)} \right) = s - c_t \geq 2,$$

so the closed point has codimension at least two in the determinantal quotient and is irrelevant for Serre condition R_1 . On the punctured spectrum, transversality identifies the singular locus with the inverse image of the matrices of rank at most $t - 2$. If $t = 1$ this locus is empty. If $t \geq 2$, its codimension in $\frac{R}{I_t(A)}$ is at least $\min\{s, c_{t-1}\} - c_t \geq 2$. Thus every codimension-one point is regular, so R_1 holds. Cohen–Macaulayness gives S_2 , and the Serre criterion [38, Lemma 10.157.4, Tag 031S] gives normality. Since the quotient is local, a normal ring is a domain. $\square$

Corollary 2.38. *Every prescribed finite matrix jet over any field admits a higher-order completion for which all determinantal ideals simultaneously have the heights and regularity properties of Theorem 2.37.*

Proof. Apply Theorem 2.13. The resulting matrix has finite left–right tangent-image codimension and is therefore finitely $\mathcal{G}_{lr}$ -determined. Apply Theorem 2.37. $\square$

3 Smooth stabilizers and Frobenius modality

In this section, we study the right action on isolated hypersurface singularities. We first prove that sufficiently high jet stabilizers are smooth. We then define the Frobenius algebra and show how its length and Loewy structure control modality, determinacy, separability, and the Milnor number.

Throughout this section, k is algebraically closed of characteristic $p > 0$. Put $\underline{x} = (x_1, \dots, x_n)$, $R = k[[\underline{x}]]$, and $\mathfrak{m} = \langle x_1, \dots, x_n \rangle$, and let $f \in \mathfrak{m}^2$. We write $\partial_i = \frac{\partial}{\partial x_i}$ and $\text{Jac}(f) = \langle \partial_1 f, \dots, \partial_n f \rangle$ for the Jacobian ideal. Throughout, $\text{Jac}(-)$ denotes the Jacobian ideal of a scalar germ, whereas $\text{JacMat}(-)$ denotes the Jacobian matrix of a vector-valued germ, as defined in Section 2. We set $\mu(f) = \dim_k \left(\frac{R}{\text{Jac}(f)} \right)$ and $\text{Tj}(f) = \dim_k \left(\frac{R}{\langle f \rangle + \text{Jac}(f)} \right)$. Thus $\mu(f)$ and $\text{Tj}(f)$ are the Milnor and Tjurina¹ numbers, respectively. We call f isolated if $\mu(f) < \infty$, equivalently if $\text{Jac}(f)$ is $\mathfrak{m}$ -primary. We reserve $\tau(-)$ exclusively for generalized test ideals in Section 6. The right group is $\mathcal{R} = \text{Aut}_k(R)$.

For $N \geq 1$ let $\mathcal{J}_N = \frac{\mathfrak{m}}{\mathfrak{m}^{N+1}}$ and let $\mathcal{R}_N$ be the algebraic group of N -jets of local coordinate changes. Its tangent space at the identity is $\left(\frac{\mathfrak{m}}{\mathfrak{m}^{N+1}} \right)^n$. For the orbit map $o_{f,N} : \mathcal{R}_N \rightarrow \mathcal{J}_N$, $\varphi \mapsto \varphi(j^N f)$, the differential at the identity and the corresponding tangent image are

$$d_{e,o_{f,N}}(v_1, \dots, v_n) = \sum_i v_i \partial_i f \pmod{\mathfrak{m}^{N+1}}, \quad (1)$$

$$\tilde{T}_{j^N f}(\mathcal{R}_N \cdot j^N f) = \frac{\mathfrak{m} \text{Jac}(f) + \mathfrak{m}^{N+1}}{\mathfrak{m}^{N+1}}. \quad (2)$$

Definition 3.1. Let an algebraic group G act on a variety X . A *Rosenlicht stratification* is a finite disjoint decomposition $X = \coprod_a X_a$ into G -invariant locally closed subvarieties

¹The Tjurina number of f is usually denoted by $\tau(f)$ (see, e.g. [13]). Since we reserve the notation $\tau(-)$ for the test ideal, we use here the notation $\text{Tj}(f)$.

for which geometric quotient morphisms $\pi_a : X_a \rightarrow Q_a$ exist. Here a *geometric quotient* means a surjective G -invariant morphism whose geometric fibers are precisely the G -orbits and whose regular functions are the invariant regular functions; the symbol Q_a denotes the corresponding quotient variety. Such stratifications exist by Rosenlicht [36, 37]. For an open neighborhood U of $x \in X$, set $G\text{-mod}(U) = \max_a \dim \pi_a(U \cap X_a)$ and define $G\text{-mod}(x)$ as the minimum of this number over all open neighborhoods U of x . The result is independent of the chosen Rosenlicht stratification [14, Definition 2.1 and Corollary A.3]. Thus $G\text{-mod}(x)$ measures the local dimension of the orbit space near x —the number of essential parameters left after quotienting by G —rather than the dimension of the orbit itself.

For an isolated hypersurface germ f , an integer N is *sufficiently large for right equivalence* if some neighborhood of $j^N f$ in $\mathcal{J}_N$ consists entirely of right N -determined germs. The *right modality* is $\text{Rmod}(f) = \mathcal{R}_N\text{-mod}(j^N f)$ for any such N ; it is independent of the choice of sufficiently large N [14, Definitions 2.1, 2.3 and Proposition 2.4].

Definition 3.2. Let (T, t_0) be a pointed algebraic variety. An *unfolding with trivial section* of f over (T, t_0) is a family $F(x, t) \in \mathcal{O}(T)[[x]]$ such that $F(x, t_0) = f$ and $F(0, t) = 0$ for every t . It is *right-complete* if every unfolding of f becomes, after passage to an étale neighborhood of its base point, fiberwise right equivalent to a pullback of F ; see [14, Definition 2.9]. A semiuniversal unfolding with trivial section has the form $F(x, t) = f(x) + \sum_{j=1}^{b_f} t_j g_j(x)$, where the classes of the g_j form a k -basis of $\frac{\mathfrak{m}}{\mathfrak{m} \text{Jac}(f)}$ and $b_f = \mu(f) + n - 1$; it is right-complete [14, Proposition 2.14].

For any unfolding F , the notation $\mathcal{R}\text{-mod}_F(f)$ denotes the local number of parameters that remain after quotienting the image of the parameter-to-jet map $t \mapsto j^N F_t$ by the $\mathcal{R}_N$ -action, computed on a Rosenlicht stratification; more precisely it is the relative Rosenlicht modality at t_0 for a sufficiently large N . It is independent of that choice [14, Definition 2.5]. For a semiuniversal unfolding with trivial section, let $\mathcal{C}_{\mu(f)}(F) = \{t \mid \mu(F_t) = \mu(f)\}$ be its Milnor-constant stratum. Following Nguyen, the *proper modality* is $\text{pmod}(f) = \dim_{t_0} \mathcal{C}_{\mu(f)}(F)$ [33, §2.2].

Remark 3.3. The notion of modality was introduced into singularity theory in the classification of simple, unimodal, and bimodal singularities by Arnold and collaborators; see [2]. The algebraic formulation used here replaces a local analytic orbit space by Rosenlicht quotients and was developed in arbitrary characteristic by Greuel–Nguyen [14]. In the complex analytic setting, Gabrielov proved that modality agrees with proper modality [12]. Nguyen established the corresponding positive-characteristic equality first for univariate germs and later for all right unimodal and bimodal germs [32, 33].

For a dominant morphism of integral varieties, we use *separable* in the usual function-field sense. Thus an orbit map is separable if the induced extension of function fields is separable; for the finite jet actions below this is equivalent to surjectivity of the differential onto the tangent space of the orbit [18, Corollary 4.2].

The external inputs used repeatedly in the hypersurface part are collected in one lemma. This also fixes the jet-order conventions used below.

Lemma 3.4 (Standard inputs for right equivalence). *Let $f \in \mathfrak{m}^2 \subseteq R = k[[x_1, \dots, x_n]]$ be isolated.*

- (i) *Every $N \geq 2\mu(f)$ is sufficiently large for right equivalence in the sense of Definition 3.1.*
- (ii) *For every unfolding F of f one has $\mathcal{R}\text{-mod}_F(f) \leq \text{Rmod}(f)$; equality holds for a right-complete unfolding. In particular, a semiuniversal unfolding with trivial section is right-complete.*
- (iii) *For an algebraic finite-jet action, the orbit map is separable if and only if its differential at the identity is surjective onto the tangent space of the orbit; equivalently, the tangent image equals the orbit tangent space.*
- (iv) *Over a perfect field, the reduction of a finite-type group scheme is a closed subgroup scheme, and every reduced finite-type group scheme is smooth.*
- (v) *If $\mathfrak{m}^{N+2} \subseteq \mathfrak{m}^2 \text{Jac}(f)$, then f is right $(2N - \text{ord}(f) + 2)$ -determined.*
- (vi) *Put*

$$M_N(f) = \frac{R}{\mathfrak{m}^N \text{Jac}(f)},$$

and regard it as a $k[[t]]$ -algebra via $t \mapsto f$. If $s = \text{ord}(f)$ and $d(f) = \text{ord}(\text{Jac}(f))$, then for every N satisfying

$$\mathfrak{m}^{\lfloor \frac{N+s+d(f)+1}{2} \rfloor} \subseteq \mathfrak{m}^2 \text{Jac}(f)$$

one has

$$f \stackrel{\mathcal{R}}{\sim} g \iff M_N(f) \cong M_N(g) \quad \text{as } k[[t]]\text{-algebras.}$$

- (vii) *There is a right equivalence $f \stackrel{\mathcal{R}}{\sim} g(x_1, \dots, x_c) + Q(x_{c+1}, \dots, x_n)$, where Q is a nonsingular quadratic form in the split variables, with the zero-variable case allowed, and $\text{Hess}(g)(0) = 0$. If $p \neq 2$, then $g \in \langle x_1, \dots, x_c \rangle^3$; if $p = 2$, the quadratic part of g is the square of a linear form. The residual germ is unique up to right equivalence. In particular, c is invariant under right equivalence, and right modality is invariant under nonsingular quadratic suspension.*

Proof. For Assertion (i), choose a jet order large enough that the Jacobian colength is computed in the corresponding truncated power-series ring. On that finite-dimensional jet space, μ is the dimension of the cokernel of a matrix whose entries depend algebraically on the jet coefficients, and is therefore upper semicontinuous. Hence, after shrinking around $j^N f$, every nearby germ g has $\mu(g) \leq \mu(f)$. Boubakri–Greuel–Markwig [6, Theorem 2.1(a) and Corollary 2.4(a)] show that an isolated g is right determined of order at most $2\mu(g) - \text{ord}(g) + 2 \leq 2\mu(g)$. Thus every $N \geq 2\mu(f)$ is sufficiently large in the sense of

Definition 3.1. The independence of right modality from the choice of a sufficiently large jet is [14, Proposition 2.4]. Assertion (ii) is [14, Proposition 2.12(ii) and Proposition 2.14]. Assertion (iii) is the standard separability criterion for algebraic group orbits; see [18, Corollary 4.2] and [13, Proposition 3.1.17]. Assertion (iv) is [38, Lemmas 39.7.6 and 39.8.4, Tags 047R and 047P]. Assertion (v) is [6, Theorem 2.1(a)], and Assertion (vi) is [19, Theorem 2.4]. Finally, Assertion (vii) is the existence-and-uniqueness form of the splitting lemma of Greuel–Pfister [15]; the invariance of right modality under nonsingular quadratic suspension follows from [14, §3]. $\square$

Remark 3.5. Over the complex numbers, the Mather–Yau theorem determines an isolated hypersurface singularity from its moduli algebra [29]. The literal analogue fails in positive characteristic; the higher Milnor-algebra form used in Lemma 3.4(vi) is the replacement proved by Greuel–Pham [19]. The splitting lemma is classical in characteristic different from two; the form used here, including characteristic two, is due to Greuel–Pfister [15].

We first study the stabilizer of a sufficiently high right jet. The argument has three steps. First, the Jacobian partials form a regular sequence. Second, exact infinitesimal stabilizers can be integrated without dividing by factorials. Third, sufficiently deep approximate stabilizers already fix the chosen jet. These three facts give smoothness of the high-jet stabilizer.

Lemma 3.6. *If f is isolated, then $\partial_1 f, \dots, \partial_n f$ is a regular sequence. Every syzygy $z = (z_1, \dots, z_n) \in R^n$ satisfying $\sum_i z_i \partial_i f = 0$ is generated by Koszul syzygies; in particular every z_i belongs to $\text{Jac}(f)$.*

Proof. The $\mathfrak{m}$ -primary ideal $\text{Jac}(f)$ has height n and is generated by n elements. The regular local ring R is Cohen–Macaulay of dimension n [38, Lemma 10.106.3, Tag 00NQ]. Hence these generators form a system of parameters and therefore a regular sequence [38, Lemma 10.104.2, Tag 02JN]. Exactness of the Koszul complex [38, Lemma 15.31.2, Tag 062F] gives the syzygy assertion, and every coordinate of a Koszul syzygy lies in $\text{Jac}(f)$. $\square$

Lemma 3.7. *For $f \in R$ and $Y = (Y_1, \dots, Y_n)$ one has*

$$f(x + Y) - f(x) - \sum_{i=1}^n (\partial_i f)(x) Y_i \in \langle Y_1, \dots, Y_n \rangle^2 R[[Y_1, \dots, Y_n]]. \quad (3)$$

This holds over an arbitrary field.

Proof. It suffices to consider a monomial x^α . The multivariable binomial formula gives

$$(x + Y)^\alpha = x^\alpha + \sum_i \alpha_i x^{\alpha - e_i} Y_i + \sum_{\substack{\beta \leq \alpha \\ |\beta| \geq 2}} \binom{\alpha}{\beta} x^{\alpha - \beta} Y^\beta.$$

The middle sum is $\sum_i \left(\frac{\partial x^\alpha}{\partial x_i} \right) Y_i$, while every term in the last sum lies in $\langle Y_1, \dots, Y_n \rangle^2$. Applying this termwise to $f = \sum_\alpha c_\alpha x^\alpha$ is legitimate in the $\langle x_1, \dots, x_n, Y_1, \dots, Y_n \rangle$ -adic topology. No division by factorials occurs. $\square$

Lemma 3.8. Assume that $f \in \mathfrak{m}^2$ is isolated and suppose $z \in \mathfrak{m}R^n$ satisfies $\sum_i z_i \partial_i f = 0$. Then there is an automorphism $\Phi_t \in \text{Aut}_{k[[t]]} k[[t, x_1, \dots, x_n]]$ with $\Phi_0 = \text{id}$, tangent vector z at $t = 0$, and $\Phi_t(f) = f$.

Proof. By Lemma 3.6, $z_i \in \text{Jac}(f) \subseteq \mathfrak{m}$. Seek $\Phi_t(x_i) = x_i + U_i(t, x)$, $U_i = tz_i + \sum_{r \geq 2} t^r u_i^{(r)}$, $u_i^{(r)} \in \text{Jac}(f)$. Assume the coefficients below order r have been chosen so that $f(x + U) - f(x) \equiv 0 \pmod{t^r}$, and let E_r be the coefficient of t^r before $u_i^{(r)}$ is chosen. In (3), every nonlinear term has at least two U -factors, hence lies in $\text{Jac}(f)^2$. The already chosen linear part has no new t^r term. Thus $E_r \in \text{Jac}(f)^2 = \sum_i (\partial_i f) \text{Jac}(f)$, so choose $u_i^{(r)} \in \text{Jac}(f)$ with $\sum_i (\partial_i f) u_i^{(r)} = -E_r$. The t -adic recursion produces $U \in t \text{Jac}(f)[[t]]^n$ with $f(x + U) = f(x)$. Its linear coefficient matrix in the x -variables is congruent to the identity modulo t , so the formal inverse-function theorem gives the required automorphism. $\square$

To control approximate stabilizers, we measure how deeply the tangent image contains powers of the maximal ideal. Since f is isolated, $\text{Jac}(f)$ is $\mathfrak{m}$ -primary, so such a containment exists. The least exponent will enter both the determinacy and the separability estimates below.

Definition 3.9. For isolated f , set $\eta(f) = \min\{e \geq 1 \mid \mathfrak{m}^e \subseteq \mathfrak{m} \text{Jac}(f)\}$. If $d = \text{ord}(f)$, then $\eta(f) \geq d$, because $\mathfrak{m} \text{Jac}(f) \subseteq \mathfrak{m}^d$.

Lemma 3.10. Let $f \in \mathfrak{m}^d$ and $w = (w_1, \dots, w_n) \in \mathfrak{m}^q R^n$, where $q \geq 1$. Then

$$f(x + tw) - f(x) - t \sum_i w_i \partial_i f \in \mathfrak{m}^{d+2(q-1)} k[[t, x]].$$

Consequently, if $\sum_i w_i \partial_i f \in \mathfrak{m}^{N+1}$ and $d + 2(q-1) \geq N+1$, then $x \mapsto x + tw$ fixes $j^N f$.

Proof. It suffices to consider a monomial x^α of total degree $s = |\alpha| \geq d$. Expanding gives $(x + tw)^\alpha = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} x^{\alpha-\beta} t^{|\beta|} w^\beta$. The terms with $|\beta| = 0$ and $|\beta| = 1$ are respectively x^α and $t \sum_i w_i \partial_i (x^\alpha)$. If $r = |\beta| \geq 2$, then $x^{\alpha-\beta} \in \mathfrak{m}^{s-r}$ and $w^\beta \in \mathfrak{m}^{rq}$, so the corresponding term has $\mathfrak{m}$ -order at least $(s-r) + rq = s + r(q-1) \geq d + 2(q-1)$. Summing over the monomials of f proves the first assertion.

Under the additional hypotheses, both the linear term and the remainder lie in $\mathfrak{m}^{N+1} k[[t, x]]$, hence $f(x + tw) \equiv f(x) \pmod{\mathfrak{m}^{N+1}}$. Finally $q \geq 1$, so the linear part in x of $x \mapsto x + tw$ has matrix $I + tA$, whose determinant has constant term 1. Thus the substitution is an automorphism over $k[[t]]$. $\square$

Theorem 3.11. Let f be isolated, $d = \text{ord}(f)$, and $\eta = \eta(f)$. For every $N \geq 2\eta - d - 1$, the scheme-theoretic stabilizer $H_N \subseteq \mathcal{R}_N$, that is, the closed subgroup scheme representing N -jet coordinate changes that fix $j^N f$, is smooth. Consequently the orbit map is separable and

$$T_{j^N f}(\mathcal{R}_N \cdot j^N f) = \frac{\mathfrak{m} \text{Jac}(f) + \mathfrak{m}^{N+1}}{\mathfrak{m}^{N+1}}.$$

Proof. Let $v \in \mathfrak{m}R^n$ represent a tangent vector to H_N , so $\delta(v) = \sum_i v_i \partial_i f \in \mathfrak{m}^{N+1}$. The numerical hypothesis and $\eta \geq d$ imply $N \geq \eta - 1$. Hence $\mathfrak{m}^{N+1} = \mathfrak{m}^{N+1-\eta} \mathfrak{m}^\eta \subseteq \mathfrak{m}^{N+1-\eta} \mathfrak{m} \text{Jac}(f) = \mathfrak{m}^{N+2-\eta} \text{Jac}(f)$. Choose $w \in \mathfrak{m}^q R^n$, where $q = N + 2 - \eta \geq 1$, such that $\delta(w) = \delta(v)$, and put $z = v - w$. Then $\delta(z) = 0$. Lemma 3.8 gives a stabilizer arc $\alpha_z(t)$ with tangent vector z . Lemma 3.10 gives a stabilizer arc $\alpha_w(t) : x \mapsto x + tw$, because $d + 2(q - 1) = d + 2(N + 1 - \eta) \geq N + 1$. Their composition $\alpha_z(t) \circ \alpha_w(t)$ is a $k[[t]]$ -valued point of the scheme-theoretic stabilizer H_N and has tangent vector $z + w = v$ modulo $\mathfrak{m}^{N+1}$. Thus every tangent vector of H_N is represented by a formal arc in H_N .

Since $\text{Spec } k[[t]]$ is reduced, every morphism $\text{Spec } k[[t]] \rightarrow H_N$ factors uniquely through the reduction $(H_N)_{\text{red}}$. Consequently every tangent vector of H_N lies in $T_e(H_N)_{\text{red}}$; the reverse inclusion is tautological, and hence

$$T_e H_N = T_e(H_N)_{\text{red}}.$$

By Lemma 3.4(iv), $(H_N)_{\text{red}}$ is smooth. The reduction has the same underlying topological space as H_N , so

$$\dim_k T_e H_N = \dim_k T_e(H_N)_{\text{red}} = \dim_e(H_N)_{\text{red}} = \dim_e H_N.$$

Thus the local ring of H_N at the identity has embedding dimension equal to its Krull dimension and is regular. Since k is perfect, regularity of this finite-type group scheme at the identity implies smoothness there; translations by group-valued points then give smoothness everywhere. Finally $\mathcal{R}_N$ is smooth, and the orbit dimension formula yields

$$\text{rank}(d_e o_{f,N}) = \dim \mathcal{R}_N - \dim H_N = \dim(\mathcal{R}_N \cdot j^N f).$$

The differential is therefore onto the tangent space of the orbit, and Lemma 3.4(iii) gives separability. $\square$

Corollary 3.12. *If $\mu(f) = \mu$, then $\mathfrak{m}^\mu \subseteq \text{Jac}(f)$, hence $\mathfrak{m}^{\mu+1} \subseteq \mathfrak{m} \text{Jac}(f)$ and $\eta(f) \leq \mu + 1$. Consequently every $N \geq 2\mu - 1$ is a separable level; in particular every $N \geq 2\mu$ is simultaneously sufficiently large for right determinacy and separability.*

Proof. The Artin local algebra $\frac{R}{\text{Jac}(f)}$ has length μ , so its maximal ideal has μ -th power zero. The rest follows from Theorem 3.11 and $d \geq 2$. $\square$

Remark 3.13. Greuel–Pham observed that contact orbit maps may be inseparable in positive characteristic, but reported that no nonseparable example was known for the right group when the hypersurface singularity is isolated [18, Section 5]. Theorem 3.11 shows that such an example cannot occur in sufficiently high jet order: every isolated hypersurface singularity has separable right orbit maps from the explicit level $N \geq 2\eta(f) - \text{ord}(f) - 1$ onward.

Corollary 3.14. *Let K be any field of positive characteristic, and let $f \in \mathfrak{m}^2 \subseteq K[[x_1, \dots, x_n]]$ have finite Milnor number. Put $d = \text{ord}(f)$ and $\eta = \min\{a \geq 1 : \mathfrak{m}^a \subseteq \mathfrak{m} \text{Jac}(f)\}$. Then the scheme-theoretic stabilizer of $j^N f$ in the right jet group is smooth over K whenever $N \geq \max\{1, 2\eta - d - 1\}$.*

Proof. Extend scalars to an algebraic closure $\overline{K}$. The containment $\mathfrak{m}^\eta \subseteq \mathfrak{m} \operatorname{Jac}(f)$ persists in $\overline{K}[[x]]$, so the extended germ remains isolated and its corresponding exponent is at most η . Its order remains d . Theorem 3.11 therefore makes the base-changed stabilizer smooth over $\overline{K}$. Stabilizers for the finite jet action commute with field extension, and smoothness descends under the faithfully flat extension $K \subseteq \overline{K}$. $\square$

We now define the Frobenius invariant used throughout the rest of the paper. The structural role of Frobenius in regular local rings goes back to the theorem of Kunz characterizing regularity by flatness of Frobenius [26]. Here we use the explicit power-series structure: the invariant records deformation directions coming from the common kernel of the derivations in characteristic p .

Let $S = R^p = k[[x_1^p, \dots, x_n^p]]$ be the image of the Frobenius endomorphism $F_R : R \rightarrow R$, $h \mapsto h^p$, and let $\mathfrak{n}_F = \langle x_1^p, \dots, x_n^p \rangle S$. Since k is perfect, S is exactly the common kernel of the derivations $\partial_1, \dots, \partial_n$.

Definition 3.15. For an isolated germ f , put

$$\begin{aligned} Q_F(f) &= S \cap \mathfrak{m} \operatorname{Jac}(f), \\ A_F(f) &= \frac{S}{Q_F(f)}, \\ \operatorname{Fmod}(f) &= \dim_k \left(\frac{\mathfrak{n}_F}{Q_F(f)} \right) = \operatorname{length}_k(A_F(f)) - 1. \end{aligned}$$

The *Frobenius type* is the $\operatorname{Aut}_k(S)$ -orbit of $Q_F(f)$.

Proposition 3.16. *The algebra $A_F(f)$, its Frobenius type, and $\operatorname{Fmod}(f)$ are invariant under right equivalence. Moreover $\operatorname{Fmod}(f) < \infty$ if and only if f is isolated.*

Proof. If $g = \varphi(f)$ with $\varphi \in \mathcal{R}$, the chain rule and invertibility of the Jacobian matrix of φ give $\operatorname{Jac}(g) = \varphi(\operatorname{Jac}(f))$. Also $\varphi(S) = S$ and $\varphi(\mathfrak{n}_F) = \mathfrak{n}_F$, since S is the common kernel of the derivations. Hence $Q_F(g) = \varphi(Q_F(f))$.

If f is isolated, choose a with $\mathfrak{m}^a \subseteq \operatorname{Jac}(f)$. For $pr \geq a + 1$, $\mathfrak{n}_F^r \subseteq \mathfrak{m}^{pr} \subseteq \mathfrak{m} \operatorname{Jac}(f)$, so $A_F(f)$ has finite length. Conversely, if $\mathfrak{n}_F^r \subseteq Q_F(f)$, then $\langle x_1^p, \dots, x_n^p \rangle^r \subseteq \mathfrak{m} \operatorname{Jac}(f)$. For a monomial x^α write $\alpha_i = pq_i + r_i$ with $0 \leq r_i < p$. If $\sum_i q_i \leq r - 1$, then $|\alpha| \leq p(r - 1) + n(p - 1)$. Hence $\mathfrak{m}^{p(r-1)+n(p-1)+1} \subseteq \langle x_1^p, \dots, x_n^p \rangle^r \subseteq \mathfrak{m} \operatorname{Jac}(f) \subseteq \operatorname{Jac}(f)$, so $\operatorname{Jac}(f)$ is $\mathfrak{m}$ -primary. This argument does not presuppose isolatedness. $\square$

For the modality comparison we use the finite-dimensional tangent algebra $M_f = \frac{R}{\mathfrak{m} \operatorname{Jac}(f)}$. The image of the nonconstant p -th powers in M_f records the Frobenius deformation directions surviving modulo $\mathfrak{m} \operatorname{Jac}(f)$. The next proposition compares those directions with proper and right modality.

Proposition 3.17. *For every isolated f , one has $\operatorname{Fmod}(f) \leq \operatorname{pmod}(f)$ and $\operatorname{Fmod}(f) \leq \operatorname{Rmod}(f)$.*

Proof. Let $d = \text{Fmod}(f)$ and choose polynomial representatives $h_1, \dots, h_d \in \mathfrak{n}_F$ whose classes form a basis of $\frac{\mathfrak{n}_F}{Q_F(f)}$. Put $V = \text{Span}_k\{h_1, \dots, h_d\}$. Then $V \cap \mathfrak{m} \text{Jac}(f) = 0$. For $g_t = f + \sum_a t_a h_a$, all first derivatives of the h_a vanish, so $\text{Jac}(g_t) = \text{Jac}(f)$ and $\mu(g_t) = \mu(f)$.

Choose N sufficiently large that $\mathfrak{m}^{N+1} \subseteq \mathfrak{m} \text{Jac}(f)$, all h_a survive in $\mathcal{J}_N$, right determinacy holds, and Theorem 3.11 applies uniformly to the family. The affine parameter space has tangent space $j^N V$, whereas the orbit tangent space at every member is $\frac{\mathfrak{m} \text{Jac}(f) + \mathfrak{m}^{N+1}}{\mathfrak{m}^{N+1}}$. These spaces meet trivially. Hence a positive-dimensional right-equivalence fiber would have a smooth point with a nonzero tangent vector in their intersection, a contradiction. Thus the general right-equivalence fiber is zero-dimensional, giving $d \leq \text{Rmod}(f)$.

The same directions occur in a semiuniversal unfolding with trivial section, and the corresponding d -plane is μ -constant. Hence $d \leq \text{pmod}(f)$. $\square$

Definition 3.18. Set

$$\varepsilon_F(f) = \text{length}_R \left(\frac{\mathfrak{m} \text{Jac}(f)}{Q_F(f)R} \right), \quad \gamma_p(f) = \dim_k \left(\frac{R}{\mathfrak{m} \text{Jac}(f) + \langle x_1^p, \dots, x_n^p \rangle} \right).$$

By the Nakayama lemma, $\gamma_p(f)$ is the minimal number of generators of $M_f = \frac{R}{\mathfrak{m} \text{Jac}(f)}$ as an $A_F(f)$ -module; this is why we call it the generator number.

Theorem 3.19. *If $d = \text{Fmod}(f)$, then*

$$p^n(d+1) = \mu(f) + n + \varepsilon_F(f). \quad (4)$$

Moreover

$$\mu(f) + n \leq \gamma_p(f)(d+1) \leq \gamma_p(f)(\text{Rmod}(f) + 1), \quad \gamma_p(f) \leq p^n. \quad (5)$$

In particular

$$\text{Rmod}(f) \geq \text{Fmod}(f) \geq \left\lceil \frac{\mu(f) + n}{p^n} \right\rceil - 1. \quad (6)$$

Proof. The ring R is free of rank p^n over S , with basis $x_1^{a_1} \cdots x_n^{a_n}$, $0 \leq a_i < p$. Hence $\dim_k \left(\frac{R}{Q_F(f)R} \right) = p^n(d+1)$. The exact sequence $0 \rightarrow \frac{\mathfrak{m} \text{Jac}(f)}{Q_F(f)R} \rightarrow \frac{R}{Q_F(f)R} \rightarrow \frac{R}{\mathfrak{m} \text{Jac}(f)} \rightarrow 0$ and $0 \rightarrow \frac{\text{Jac}(f)}{\mathfrak{m} \text{Jac}(f)} \rightarrow \frac{R}{\mathfrak{m} \text{Jac}(f)} \rightarrow \frac{R}{\text{Jac}(f)} \rightarrow 0$ give (4), since $\dim_k \left(\frac{\text{Jac}(f)}{\mathfrak{m} \text{Jac}(f)} \right) = n$ by Lemma 3.6.

Furthermore $Q_F(f) = \text{Ann}_S(M_f)$, because an element of S annihilates M_f exactly when it kills $[1]$. Thus M_f is a finite $A_F(f)$ -module, minimally generated by $\gamma_p(f)$ elements. Nakayama gives a surjection $A_F(f)^{\gamma_p(f)} \twoheadrightarrow M_f$, whence $\mu(f) + n \leq \gamma_p(f)(d+1)$. Finally $\frac{M_f}{\mathfrak{n}_F M_f}$ is a quotient of $\frac{R}{\langle x_1^p, \dots, x_n^p \rangle}$, so $\gamma_p(f) \leq p^n$. $\square$

Corollary 3.20. *Equality $\mu(f) + n = \gamma_p(f)(\text{Fmod}(f) + 1)$ holds exactly when M_f is free over $A_F(f)$ of rank $\gamma_p(f)$. Equality $\mu(f) + n = p^n(\text{Fmod}(f) + 1)$ holds exactly when $\mathfrak{m} \text{Jac}(f) = Q_F(f)R$.*

Proof. The first assertion follows because the surjection $A_F(f)^{\gamma_p(f)} \twoheadrightarrow M_f$ is an isomorphism exactly when the finite lengths agree. The second follows from the kernel $\frac{\mathfrak{m} \text{Jac}(f)}{Q_F(f)R}$ of $\frac{R}{Q_F(f)R} \twoheadrightarrow \frac{R}{\mathfrak{m} \text{Jac}(f)}$. $\square$

Proposition 3.21. *The Frobenius map on M_f satisfies $\text{ima } F \cong A_F(f)$ and $\text{Fmod}(f) + 1 = \dim_k \text{ima } F$. If $b_e(f) = \dim_k \text{ima } F^e$, then $\text{Rmod}(f) \geq b_e(f) - 1$, $b_1(f) \geq b_2(f) \geq \cdots \geq 1$, and $b_e(f) = 1$ for $e \gg 0$.*

Proof. The image of Frobenius on R is S , and the kernel of $S \rightarrow M_f$ is $Q_F(f)$. Replacing p by p^e in the argument of Proposition 3.17 gives $\text{Rmod}(f) \geq b_e(f) - 1$. The images are nested, and a sufficiently high Frobenius power kills the maximal ideal of the Artin algebra M_f . $\square$

Iterating Frobenius gives a nested sequence of finite-dimensional quotients and corresponding numerical bounds. For $e \geq 1$, set $S_e = k[[x_1^{p^e}, \dots, x_n^{p^e}]]$, $\mathfrak{n}_e = \langle x_1^{p^e}, \dots, x_n^{p^e} \rangle S_e$, $Q_e = S_e \cap \mathfrak{m} \text{Jac}(f)$, $A_e = \frac{S_e}{Q_e}$, and $b_e = \dim_k A_e$. Also set $\gamma_e(f) = \dim_k \left(\frac{M_f}{\mathfrak{n}_e M_f} \right)$ and $\varepsilon_e(f) = \text{length}_R \left(\frac{\mathfrak{m} \text{Jac}(f)}{Q_e R} \right)$.

Theorem 3.22. *For every $e \geq 1$:*

- (i) $A_e \cong \text{ima } F^e$ and $p^{en} b_e = \mu(f) + n + \varepsilon_e(f)$.
- (ii) $\mu(f) + n \leq \gamma_e(f) b_e$ and $\gamma_e(f) \leq p^{en}$.
- (iii) $\text{Rmod}(f) \geq b_e - 1$ and $\text{pmod}(f) \geq b_e - 1$.
- (iv) $b_1 \geq b_2 \geq \cdots \geq 1$, and $b_e = 1$ for $e \gg 0$.

Proof. The proof is identical to that of Theorem 3.19, replacing S by S_e and p^n by p^{en} . The nonconstant elements of S_e have zero first derivatives, so the same Frobenius-slice argument gives both modality inequalities. $\square$

Definition 3.23. The *Loewy length* of a local Artin algebra $(A, \mathfrak{q})$ is the least integer $\ell \geq 1$ such that $\mathfrak{q}^\ell = 0$. We denote the Loewy length of $A_F(f)$ by $\ell_F(f) = \min\{\ell \geq 1 \mid \mathfrak{n}_F^\ell \subseteq Q_F(f)\}$ and put $B_F(f) = p(\ell_F(f) - 1) + n(p - 1)$.

Lemma 3.24. *For isolated f , we have $\mathfrak{m}^{B_F(f)+1} \subseteq \langle x_1^p, \dots, x_n^p \rangle^{\ell_F(f)} \subseteq Q_F(f)R \subseteq \mathfrak{m} \text{Jac}(f)$. Consequently, we get that $\eta(f) \leq B_F(f) + 1$.*

Proof. The middle containments are immediate. Write a monomial exponent as $\alpha_i = pq_i + r_i$, $0 \leq r_i < p$. If $\sum_i q_i \leq \ell_F(f) - 1$, then $|\alpha| \leq p(\ell_F(f) - 1) + n(p - 1) = B_F(f)$. Thus every monomial of degree at least $B_F(f) + 1$ lies in $\langle x_1^p, \dots, x_n^p \rangle^{\ell_F(f)}$. $\square$

The Loewy estimate can now be inserted simultaneously into the determinacy, separability, and Mather–Yau bounds. By a *right classification radius* we mean an integer that is simultaneously a right-determinacy level, a separability level for the right jet orbit, and a level at which the higher Milnor algebra is a complete right-equivalence invariant.

Theorem 3.25. *Let $f \in \mathfrak{m}^2$ be isolated and set $N_\eta(f) = 2\eta(f) - \text{ord}(f)$. Then*

- (i) *f is right $N_\eta(f)$ -determined;*
- (ii) *every right orbit map at level $N \geq N_\eta(f) - 1$ is separable;*
- (iii) *the $k[[t]]$ -algebra*

$$M_{N_\eta(f)}(f) = \frac{R}{\mathfrak{m}^{N_\eta(f)} \text{Jac}(f)}$$

is a complete right-equivalence invariant.

Consequently,

$$2B_F(f) - \text{ord}(f) + 2$$

is also a right classification radius.

Proof. Since $\mathfrak{m}^{\eta(f)} \subseteq \mathfrak{m} \text{Jac}(f)$, we have that $\mathfrak{m}^{\eta(f)+1} \subseteq \mathfrak{m}^2 \text{Jac}(f)$, and so from Lemma 3.4(v) we have that f is right $(2\eta - d)$ -determined. In addition, Assertion (ii) is Theorem 3.11. Finally, for $d_J = \text{ord}(\text{Jac}(f)) \geq 1$ we have that

$$\left\lfloor \frac{N_\eta(f) + \text{ord}(f) + d_J + 1}{2} \right\rfloor = \eta(f) + \left\lfloor \frac{d_J + 1}{2} \right\rfloor \geq \eta(f) + 1.$$

Thus the hypothesis of Lemma 3.4(vi) holds, proving (iii). Finally $\eta \leq B_F(f) + 1$ gives the stated Frobenius-Loewy radius. $\square$

Corollary 3.26. *If $r = \text{Rmod}(f)$, then*

$$\langle x_1^p, \dots, x_n^p \rangle^{r+1} \subseteq \mathfrak{m} \text{Jac}(f), \quad \eta(f) \leq pr + n(p-1) + 1.$$

Proof. Since $\text{Fmod}(f) \leq r$, the maximal ideal of $A_F(f)$ has $(r+1)$ -st power zero. Apply Lemma 3.24. $\square$

We first record the Frobenius algebras of the first few lengths. This gives a concrete picture of the invariant through Frobenius modality three.

Write $y_i = x_i^p$, so $S = k[[y_1, \dots, y_n]]$. Every automorphism of S lifts to one of R : take coefficientwise p -th roots of the images of the y_i . Thus Frobenius type is a right-coordinate invariant.

If $d = \text{Fmod}(f)$, then $Q_F(f)$ has colength $d+1$ and contains $\mathfrak{n}_F^{d+1}$. Hence fixed- d Frobenius-type classification is the finite dimensional $\text{Aut}_k\left(\frac{S}{\mathfrak{n}_F^{d+1}}\right)$ -orbit problem for the colength- $(d+1)$ ideals realized as $Q_F(f)$.

Proposition 3.27. *Up to $\text{Aut}_k(S)$, every colength- $(d+1)$ ideal $Q \subseteq S$ for $d \leq 3$ has one of the following forms, with undisplayed variables killed linearly:*

d	Q
0	$\langle y_1, \dots, y_n \rangle,$
1	$\langle y_2, \dots, y_n, y_1^2 \rangle,$
2	$\langle y_2, \dots, y_n, y_1^3 \rangle,$ $\langle y_3, \dots, y_n \rangle + \langle y_1, y_2 \rangle^2,$
3	$\langle y_2, \dots, y_n, y_1^4 \rangle,$ $\langle y_3, \dots, y_n, y_1 y_2, y_2^2, y_1^3 \rangle,$ $\langle y_3, \dots, y_n, y_1^2, y_2^2 \rangle,$ $\langle y_4, \dots, y_n \rangle + \langle y_1, y_2, y_3 \rangle^2.$

In characteristic 2 there is in addition the nonalternating $(1, 2, 1)$ type

$$\langle y_3, \dots, y_n, y_1 y_2, y_1^2 + y_2^2 \rangle.$$

Only types realized as $S \cap \mathfrak{m} \text{Jac}(f)$ occur for hypersurface germs.

Proof. The possible Hilbert functions of a local Artin algebra of length at most four are (1) , $(1, 1)$, $(1, 1, 1)$, $(1, 2)$, $(1, 1, 1, 1)$, $(1, 2, 1)$, and $(1, 3)$. All cases except $(1, 2, 1)$ follow by choosing a generator in each nonzero graded piece and lifting a basis of the embedding space. In the remaining case multiplication induces, after choosing a generator of the one-dimensional vector space $\mathfrak{q}^2$, a nonzero symmetric bilinear form on the two-dimensional vector space $\mathfrak{q}^2$. Changing generators acts by congruence and rescaling the socle rescales the form. Over an algebraically closed field of characteristic different from 2, the form has rank 1 or 2. In characteristic 2, the nondegenerate case has two congruence types, alternating and nonalternating. Writing normal forms for these bilinear forms gives exactly the displayed ideals. $\square$

Proposition 3.28. *For an isolated $f \in \mathfrak{m}^2$, the following are equivalent:*

- (i) $\text{Fmod}(f) = 0$;
- (ii) $Q_F(f) = \mathfrak{n}_F$;
- (iii) $x_i^p \in \mathfrak{m} \text{Jac}(f)$ for every i ;
- (iv) Frobenius on $M_f = \frac{R}{\mathfrak{m} \text{Jac}(f)}$ kills its maximal ideal.

If these conditions hold, then

$$\mathfrak{m}^{n(p-1)+1} \subseteq \mathfrak{m} \text{Jac}(f), \quad \mu(f) + n \leq p^n,$$

and f is right $2n(p-1)$ -determined.

Proof. The local algebra $A_F(f) = \frac{S}{Q_F(f)}$ has length one exactly when $Q_F(f) = \mathfrak{n}_F$. Since $\mathfrak{n}_F = \langle x_1^p, \dots, x_n^p \rangle S$, this is equivalent to (iii). Because k is perfect, (iii) is equivalent to (iv). The remaining assertions follow from Lemma 3.24, Theorem 3.25, and Theorem 3.19. $\square$

Put $B = n(p-1)$, $N = 2B$, and $\mathcal{J}_N^{(2)} = \frac{\mathfrak{m}^2}{\mathfrak{m}^{N+1}}$. Let $\mathcal{Z}_{n,p} \subseteq \mathcal{J}_N^{(2)}$ be the set of jets u satisfying $x_i^p \in \mathfrak{m} \text{Jac}(\tilde{u}) + \mathfrak{m}^{N+1}$ for every i , where $\tilde{u}$ is any lift of u . The definition is independent of the lift, since changing a lift by $\mathfrak{m}^{N+1}$ changes its partial derivatives by $\mathfrak{m}^N$ and hence $\mathfrak{m} \text{Jac}(\tilde{u})$ by terms in $\mathfrak{m}^{N+1}$.

Theorem 3.29. *The jet map induces a bijection*

$$\frac{\{f \in \mathfrak{m}^2 \mid \text{Fmod}(f) = 0\}}{\underset{\sim}{\mathcal{R}}} \longrightarrow \frac{\mathcal{Z}_{n,p}}{\mathcal{R}_N}, \quad N = 2n(p-1).$$

Equivalently, every such germ is right equivalent to a polynomial of degree at most N , and two such germs are right equivalent exactly when their N -jets lie in the same $\mathcal{R}_N$ -orbit.

Proof. The forward implication follows from Proposition 3.28. Conversely let $u \in \mathcal{Z}_{n,p}$ and let f be a lift. Since $\mathfrak{m}^{B+1} \subseteq \langle x_1^p, \dots, x_n^p \rangle$, the defining condition gives $\mathfrak{m}^{B+1} \subseteq \mathfrak{m} \text{Jac}(f) + \mathfrak{m}^{2B+1}$. Set $L = \frac{\mathfrak{m}^{B+1} + \mathfrak{m} \text{Jac}(f)}{\mathfrak{m} \text{Jac}(f)}$. Because $\mathfrak{m}^{2B+1} = \mathfrak{m}^B \mathfrak{m}^{B+1}$, the preceding inclusion implies $L \subseteq \mathfrak{m}^B L \subseteq \mathfrak{m} L$. Since also $\mathfrak{m} L \subseteq L$, we obtain $L = \mathfrak{m} L$, and the Nakayama lemma gives $L = 0$. Hence $\mathfrak{m}^{B+1} \subseteq \mathfrak{m} \text{Jac}(f)$ and therefore $\mathfrak{m}^{2B+1} \subseteq \mathfrak{m} \text{Jac}(f)$. The congruences defining $\mathcal{Z}_{n,p}$ consequently become exact: $x_i^p \in \mathfrak{m} \text{Jac}(f)$. Thus $\text{Fmod}(f) = 0$. Theorem 3.25 makes every such germ right N -determined, so right equivalence is exactly the $\mathcal{R}_N$ -orbit relation on N -jets. $\square$

Corollary 3.30. (i) *If $n = 1$, the isolated germs f with $\text{Fmod}(f) = 0$ are exactly x^d , up to right equivalence, with $2 \leq d \leq p-1$.*

(ii) *If $p = 2$, an isolated germ f has $\text{Fmod}(f) = 0$ if and only if n is even and $f \underset{\sim}{\mathcal{R}} x_1 x_2 + x_3 x_4 + \dots + x_{n-1} x_n$.*

Proof. For $n = 1$, write $f = \sum_{j \geq 2} a_j x^j$. Since f is isolated, there is a least d such that $a_d \neq 0$ and $p \nmid d$, and then $\text{Jac}(f) = \langle x^{d-1} \rangle$ up to a unit, so $\mathfrak{m} \text{Jac}(f) = \langle x^d \rangle$. The image of $k[[x^p]]$ in $\frac{R}{\mathfrak{m} \text{Jac}(f)}$ has basis $1, x^p, x^{2p}, \dots, x^{\lfloor \frac{d-1}{p} \rfloor p}$. Hence $\text{Fmod}(f) = \lfloor \frac{d-1}{p} \rfloor$. Thus $\text{Fmod}(f) = 0$ forces $d \leq p$, and since $p \nmid d$ we have $d \leq p-1$. There is then no positive multiple of p below d , so $d = \text{ord}(f)$ and $f = x^d u(x)$ with $u(0) \neq 0$. Because $p \nmid d$, the unit u has a d -th root in $k[[x]]$, giving $f \underset{\sim}{\mathcal{R}} x^d$.

For $p = 2$, apply Lemma 3.4(vii): $f \underset{\sim}{\mathcal{R}} g(x_1, \dots, x_c) + q$. Put $\mathfrak{m}_g = \langle x_1, \dots, x_c \rangle \subseteq k[[x_1, \dots, x_c]]$. Suspension invariance gives $\text{Fmod}(g) = 0$. If $c > 0$, then $\text{Jac}(g) \subseteq \mathfrak{m}_g^2$, so $\mathfrak{m}_g \text{Jac}(g) \subseteq \mathfrak{m}_g^3$, contradicting $x_i^2 \in \mathfrak{m}_g \text{Jac}(g)$. Thus $c = 0$, and the remaining nonsingular quadratic form is hyperbolic over the algebraically closed field k . $\square$

Remark 3.31. For $p \geq 5$, isolated homogeneous cubics f in three variables have $\text{Fmod}(f) = 0$ but may have positive right modality. Indeed their three quadratic partial derivatives form a regular sequence, so the Jacobian algebra has socle degree 3, hence $\mathfrak{m}^4 \subseteq \text{Jac}(f)$ and $\mathfrak{m}^5 \subseteq \mathfrak{m} \text{Jac}(f)$. Since $p \geq 5$, one has $x_i^p \in \mathfrak{m} \text{Jac}(f)$. Thus $\text{Fmod}(f) = 0$ is not equivalent to right simplicity.

Finally, we compare the Frobenius directions with more general deformation directions that preserve the Jacobian ideal. This gives a decreasing persistence filtration between proper modality, right modality, and Frobenius modality.

For $r \geq 0$ define

$$\mathcal{P}_r(f) = \{h \in \mathfrak{m} \mid \partial_i h \in \mathfrak{m}^r \text{ Jac}(f) \text{ for every } i\}, \quad d_r(f) = \dim_k \left(\frac{\mathcal{P}_r(f) + \mathfrak{m} \text{ Jac}(f)}{\mathfrak{m} \text{ Jac}(f)} \right).$$

Thus $\mathcal{P}_r(f)$ consists of perturbation directions whose gradient is r steps deep in the Jacobian ideal, and $d_r(f)$ counts the directions that survive modulo the right tangent image $\mathfrak{m} \text{ Jac}(f)$; the decreasing sequence $d_r(f)$ is the persistence profile used below.

Proposition 3.32. *For every isolated f ,*

$$\text{Rmod}(f) \geq d_0(f) \geq d_1(f) \geq \cdots \geq \text{Fmod}(f), \quad d_0(f) \leq \text{pmod}(f).$$

If $d(f) = \text{ord}(\text{Jac}(f))$ and $B_F = B_F(f)$, then

$$d_r(f) = \text{Fmod}(f) \quad r \geq \max\{0, B_F - d(f)\}.$$

Proof. The inclusions $\mathcal{P}_{r+1} \subseteq \mathcal{P}_r$ and $S \cap \mathfrak{m} \subseteq \mathcal{P}_r$ give monotonicity and the lower Frobenius bound. Choose independent classes represented by $h_1, \dots, h_e \in \mathcal{P}_r$. Write the gradient of h_a as $A_a(\partial_1 f, \dots, \partial_n f)^\top$ with $A_a \in M_n(\mathfrak{m}^r)$. For $g_t = f + \sum_a t_a h_a$, its gradient is obtained from $(\partial_1 f, \dots, \partial_n f)^\top$ by multiplication with $I + \sum_a t_a A_a$. This matrix is invertible on a neighborhood of the origin in parameter space, so $\text{Jac}(g_t) = \text{Jac}(f)$. The same finite-jet transversality argument as in Proposition 3.17 gives $e \leq \text{Rmod}(f)$, and for $r = 0$ the family is μ -constant, giving $e \leq \text{pmod}(f)$.

If $r + d(f) \geq B_F$, then every partial derivative of $h \in \mathcal{P}_r$ lies in $\mathfrak{m}^{B_F}$. Consider a monomial cx^α occurring in h with total degree at most B_F . If some α_i is not divisible by p , then its image under ∂_i is a nonzero monomial of degree $< B_F$; distinct monomials of h cannot cancel after applying the same ∂_i . Thus every exponent of every such monomial is divisible by p . Since k is perfect, the truncation of h through degree B_F lies in S . Therefore $h = s + u$ with $s \in S \cap \mathfrak{m}$ and $u \in \mathfrak{m}^{B_F+1}$. By Lemma 3.24, $u \in \mathfrak{m} \text{ Jac}(f)$. Thus every surviving class is represented by a Frobenius direction. $\square$

4 Splitting, boundedness, and proper modality

In this section, we combine the Frobenius bounds with Lemma 3.4(vii). We first bound the essential corank, Milnor number, determinacy order, and multiplicity in terms of the right modality. We then study the Milnor-constant stratum of a semiuniversal unfolding and prove that proper modality equals right modality.

Let $f \stackrel{\mathcal{R}}{\sim} g(x_1, \dots, x_c) + Q(z_1, \dots, z_{n-c})$ be the splitting-lemma decomposition, where Q is nonsingular and g is the residual germ, and write $\mathfrak{m}_x = \langle x_1, \dots, x_c \rangle \subseteq k[[x_1, \dots, x_c]]$. We call $c = c(f)$ the *essential corank*; equivalently, it is the number of variables that remain

after all nonsingular quadratic summands have been split off. Two isolated hypersurface germs are *stably right equivalent* if, after adjoining finitely many variables and adding nonsingular quadratic forms in the new variables, the resulting germs are right equivalent. By Lemma 3.4(vii), the stable right-equivalence class is represented by the residual germ.

Proposition 4.1. *With this notation, $A_F(f) \cong A_F(g)$, $\text{Fmod}(f) = \text{Fmod}(g)$, $\ell_F(f) = \ell_F(g)$, and $\mu(f) = \mu(g)$.*

Proof. The partial derivatives of the nonsingular quadratic block generate the quadratic variables. Every nonconstant p -th-power monomial involving one of those variables therefore belongs to $\mathfrak{m} \text{Jac}(f)$, while setting the quadratic variables equal to zero maps $\mathfrak{m} \text{Jac}(f)$ onto $\mathfrak{m} \text{Jac}(g)$. $\square$

Applying Theorem 3.19 to the residual germ and using the suspension invariance in Proposition 4.1 gives a bound involving only the residual number of variables. In particular, the variables belonging to the nonsingular quadratic summand do not occur in the exponent. We obtain

$$\mu(f) + c(f) \leq p^{c(f)} (\text{Fmod}(f) + 1). \quad (7)$$

Lemma 4.2. *Assume $p = 2$, and let g be an isolated residual germ in c variables. Then we have that $\text{Rmod}(g) \geq c + \binom{c}{3}$.*

Proof. The assertion is immediate for $c = 0$. Assume $c > 0$. Let $V_2^{\text{sq}} = \text{Span}_k\{x_1^2, \dots, x_c^2\}$ and $V_3 = k[x_1, \dots, x_c]_3$. Since $\text{Hess}(g)(0) = 0$, its quadratic part belongs to V_2^{sq} . Consider the unfolding $g + h_2 + h_3$ for $(h_2, h_3) \in V_2^{\text{sq}} \oplus V_3$. Its parameter-to-3-jet map is injective, and the parameter space has dimension $c + \binom{c+2}{3}$.

Write a coordinate change as $\varphi(x) = Ax + u(x)$ with $u \in \mathfrak{m}^2 k[[x]]^c$. In characteristic two, substituting φ into a square quadratic form produces no cubic contribution from u . Substitution into a cubic form affects its cubic part only through A . Consequently the action on the 3-jets in this family factors through GL_c .

Every right-equivalence fiber in the parameter space is therefore contained in a GL_c -orbit at the 3-jet level, and has dimension at most c^2 . After restricting to a neighborhood of the origin where the fibers are isolated, the relative modality is at least $c + \binom{c+2}{3} - c^2 = c + \binom{c}{3}$. The modality of an unfolding is at most the right modality of its central germ, proving the assertion. $\square$

For $D \geq 0$ define

$$C_{p,D} = \begin{cases} \max \{c \in \mathbb{N} \mid \binom{c}{3} \leq D\}, & p \neq 2, \\ \max \{c \in \mathbb{Z}_{\geq 0} : c + \binom{c}{3} \leq D\}, & p = 2. \end{cases}$$

Put

$$B_{p,D} = pD + C_{p,D}(p-1), \quad N_{p,D} = 2B_{p,D}, \quad \hat{N}_{p,D} = \max\{2, N_{p,D}\}.$$

Theorem 4.3. *Fix $p > 0$ and $D \geq 0$, and let f be isolated with $\text{Rmod}(f) \leq D$. After splitting off a nonsingular quadratic block, the residual germ satisfies:*

- (i) it has at most $C_{p,D}$ variables;
- (ii) its Milnor number is at most $p^{C_{p,D}}(D+1) - C_{p,D}$;
- (iii) its right classification radius is at most $N_{p,D}$;
- (iv) if it is nonzero, its multiplicity is at most $p(D+1)$.

Proof. For $p \neq 2$, the residual germ has order at least three, and [14, Proposition 2.18] gives $\text{Rmod}(g) \geq \binom{c+\text{ord}(g)-1}{\text{ord}(g)} - c^2 \geq \binom{c}{3}$. For $p = 2$ use Lemma 4.2. Thus $c \leq C_{p,D}$ in all characteristics.

The Milnor bound follows from (7) and $\text{Fmod}(g) \leq \text{Rmod}(g) \leq D$. Theorem 3.25, applied in c variables, gives the radius $2pD + 2c(p-1) \leq N_{p,D}$.

Finally, Corollary 4.8 applied to g gives

$$D+1 \geq \left(c + \left\lceil \frac{\text{ord}(g)}{p} \right\rceil - 1 \right)^c.$$

For $c > 0$, the right-hand side is at least $\left\lceil \frac{\text{ord}(g)}{p} \right\rceil$, proving $\text{ord}(g) \leq p(D+1)$. $\square$

The residual determinacy estimate also gives a determinacy order for the original germ, not only for its stable class. The point is that the partial derivatives of the nonsingular quadratic block generate all split variables. Hence every sufficiently high-order monomial involving a split variable already lies in the full Jacobian contribution, and the residual containment extends to the original power-series ring.

Corollary 4.4. *Fix $p > 0$ and $D \geq 0$. Every isolated hypersurface germ f with $\text{Rmod}(f) \leq D$ is right $\hat{N}_{p,D}$ -determined, independently of the number of ambient variables.*

Proof. Write $f \stackrel{\mathcal{R}}{\sim} g(x_1, \dots, x_c) + Q(z_1, \dots, z_{n-c})$ as in Lemma 3.4(vii). If $c = 0$, the germ is a nonsingular quadratic form and is right 2-determined. Assume $c > 0$ and put $B = p(\ell_F(f) - 1) + c(p-1) = B_F(g)$. Since $\ell_F(f) - 1 \leq \text{Fmod}(f) \leq D$ and $c \leq C_{p,D}$, one has $B \leq B_{p,D}$. Lemma 3.24, applied to the residual germ, gives $\mathfrak{m}_x^{B+2} \subseteq \mathfrak{m}_x^2 \text{Jac}(g)$. On the other hand the partial derivatives of the nonsingular quadratic form Q generate the ideal $\langle z_1, \dots, z_{n-c} \rangle$. Since $B \geq 1$, every monomial of total degree at least $B+2$ that involves a split variable belongs to $\mathfrak{m}^2 \langle z_1, \dots, z_{n-c} \rangle$. Therefore, in the full power-series ring,

$$\mathfrak{m}^{B+2} \subseteq \mathfrak{m}^2 \text{Jac}(f).$$

Lemma 3.4(v) shows that f is right $(2B - \text{ord}(f) + 2)$ -determined, hence right $2B$ -determined because $\text{ord}(f) \geq 2$. Since $2B \leq N_{p,D} \leq \hat{N}_{p,D}$, the assertion follows. $\square$

Corollary 4.5. *Let $r = \text{Rmod}(f)$. Then*

$$\mu(f) \leq p^{2+\sqrt[3]{6r}}(r+1).$$

Consequently, for fixed p , any sequence (f_j) of isolated hypersurface singularities with $\mu(f_j) \rightarrow \infty$ has $\text{Rmod}(f_j) \rightarrow \infty$, even if the ambient dimension is allowed to tend to infinity.

Proof. The definition of $C_{p,r}$ and Lemma 4.2 imply $\binom{c}{3} \leq r$ for the residual essential corank in every characteristic. Hence $c(c-1)(c-2) \leq 6r$ and therefore $c \leq 2 + \sqrt[3]{6r}$. Apply (7) and $\text{Fmod}(f) \leq r$. $\square$

Corollary 4.6. *Fix $p > 0$. If $\{f_j\}_{j=1}^\infty$ is any sequence of isolated hypersurface singularities in characteristic p with $\mu(f_j) \rightarrow \infty$, then $\text{Rmod}(f_j) \rightarrow \infty$.*

Proof. This is immediate from Corollary 4.5. $\square$

Remark 4.7. Nguyen proved that $\mu(f) \leq 4p$ for singularities of right modality at most two [33, Corollary 3.7]. Corollary 4.5 gives an analogous bounded-Milnor-number statement in every finite modality and, moreover, makes the bound independent of the ambient dimension. This is compatible with the broader finiteness phenomenon conjectured by Greuel–Nguyen for singularities with an arbitrary number of moduli [14].

Corollary 4.8. *If $r = \text{ord}(f)$, then*

$$\text{Fmod}(f) \geq \left(n + \left\lceil \frac{r}{p} \right\rceil - 1 \right) - 1.$$

Proof. Since $\mathfrak{m} \text{Jac}(f) \subseteq \mathfrak{m}^r$, $Q_F(f) \subseteq S \cap \mathfrak{m}^r = \mathfrak{n}_F^{\left\lceil \frac{r}{p} \right\rceil}$. Compare colengths. $\square$

Proposition 4.9. *Let $f = \sum_{i=1}^n x_i^{a_i}$ with $p \nmid a_i$, and put $b_i = a_i - 1$. Then*

$$\text{Fmod}(f) + 1 = \prod_{i=1}^n \left\lceil \frac{b_i}{p} \right\rceil + \#\{i \mid p \mid b_i\}.$$

Proof. Up to units, $\text{Jac}(f) = \langle x_1^{b_1}, \dots, x_n^{b_n} \rangle$. The monomials outside $\mathfrak{m} \text{Jac}(f)$ are those with every exponent strictly below the corresponding b_i , together with the exceptional pure monomials $x_i^{b_i}$. Intersecting this set with the p -lattice gives the stated count. $\square$

We now study proper modality. We realize the Milnor number as a local fiber length in a finite flat family and compare the corresponding length strata with Rosenlicht strata. Recall Definition 3.2. We use the standard semiuniversal unfolding with trivial section $F(x, t) = f(x) + \sum_{j=1}^{b_f} t_j g_j(x)$, where $b_f = \mu(f) + n - 1$ and the classes of the g_j form a basis of $\frac{\mathfrak{m}}{\mathfrak{m} \text{Jac}(f)}$. We write $\mathcal{C}_{\mu(f)}(F) = \{t \mid \mu(F_t) = \mu(f)\}$ for its Milnor-constant stratum near the origin.

Theorem 4.10. *For every isolated f , one has $\text{pmod}(f) \leq \text{Rmod}(f)$.*

Proof. Put $\mu = \mu(f)$ and choose an irreducible component $C \subseteq \mathcal{C}_{\mu}(F)$ through 0 of dimension $\text{pmod}(f)$. Take $N \geq 2\mu$ sufficiently large. By Corollary 3.12, every μ -constant member has separable N -jet orbit, and the orbit tangent space has constant codimension $\mu + n - 1$.

At the origin the semiuniversal parameter space is complementary to this tangent space; hence transversality persists on a dense open subset of C .

A positive-dimensional right-equivalence fiber would have a smooth point with a nonzero tangent vector lying both in the parameter tangent space and in the orbit tangent space. This contradicts transversality. Hence the general right-equivalence fiber on C is zero-dimensional, and right-completeness gives $\text{Rmod}(f) \geq \dim C = \text{pmod}(f)$. $\square$

Put $M = \mu(f)$. By Lemma 3.4(v), we may replace f by a right-equivalent polynomial representative. Choose polynomials $h_1, \dots, h_{M-1} \in \mathfrak{m}^2$ so that $x_1, \dots, x_n, h_1, \dots, h_{M-1}$ represent a basis of $\frac{\mathfrak{m}}{\mathfrak{m} \text{Jac}(f)}$, and set $H(x, t) = f(x) + \sum_{j=1}^{M-1} t_j h_j(x)$. Then $F(x; a, t) = H(x, t) + \sum_i a_i x_i$ is the standard semiuniversal unfolding with trivial section: the classes of $x_1, \dots, x_n, h_1, \dots, h_{M-1}$ form a basis of $\frac{\mathfrak{m}}{\mathfrak{m} \text{Jac}(f)}$, the tangent space of the deformation functor with section. Eliminating the a_i from the critical equations gives

$$\Pi : \mathbb{A}_{u,t}^{n+M-1} \longrightarrow \mathbb{A}_{a,t}^{n+M-1}, \quad (u, t) \longmapsto (-\partial_1 H(u, t), \dots, -\partial_n H(u, t), t).$$

For $c = (u, t)$ define $G_c(y) = H(u+y, t) - H(u, t) - \sum_i \partial_i H(u, t) y_i$. Then $G_c \in \langle y_1, \dots, y_n \rangle^2$, and the completed local ring of the fiber of Π at c is $\frac{k[[y]]}{\text{Jac}_y(G_c)}$, where $\text{Jac}_y(G_c) = \langle \frac{\partial G_c}{\partial y_1}, \dots, \frac{\partial G_c}{\partial y_n} \rangle$. Thus the local scheme-theoretic fiber length is $\mu(G_c)$.

Let $c_0 = (0, 0)$ and $b_0 = \Pi(c_0)$. Since f is isolated, c_0 is an isolated point of the fiber over b_0 . By [38, Lemma 37.41.1, Tag 02LK], after an étale base change around b_0 there is an open neighborhood of c_0 that is finite over the new base. Shrinking this neighborhood and the base, we may moreover arrange that the chosen neighborhood contains no other point of the special fiber.

Let $c_0 = (0, 0)$ and $b_0 = \Pi(c_0)$. Since f is isolated, c_0 is an isolated point of the fiber over b_0 . By [38, Lemma 37.41.1, Tag 02LK], after an étale base change around b_0 there is an open neighborhood of c_0 finite over the new base. Shrink so that this neighborhood contains no other point of the special fiber, and replace it by the connected component containing c_0 . Because it is étale over the smooth (u, t) -space, this source is regular; a connected regular scheme is integral. Replace the target by the connected component containing the image of c_0 and shrink once more. The resulting source and target are smooth integral varieties of the same dimension, and the finite map between them is surjective. Since the source local rings are Cohen–Macaulay and the target local rings are regular, miracle flatness [38, Lemma 10.128.1, Tag 00R4] makes the map flat.

We have obtained a finite flat surjective map

$$\pi : X \longrightarrow Y.$$

By construction, the special fiber of this chosen local factor is supported only at the distinguished point, and its completed local algebra is $\frac{k[[y]]}{\text{Jac}(f)}$, of length M . Hence the finite flat rank of π is exactly M . Write $x_0 \in X$ for the distinguished point. We call π the

critical cover associated with the chosen semiuniversal unfolding; its points record critical points of nearby fibers together with their deformation parameters. For a closed point $x \in X$, put $\lambda(x) = \dim_k \mathcal{O}_{\pi^{-1}(\pi(x)), x}$. Thus $\lambda(x) = \mu(G_x)$. For an irreducible subvariety, its generic local length means the common value of λ on a dense open subset of its closed points.

Let $\iota = [k(X) : k(Y)]_{\text{insep}}$ be the inseparable degree of the critical cover. For a closed point $x \in X$, let $e_\pi(x)$ denote the Gaffney–Lazarsfeld local degree: after strict henselization at $y = \pi(x)$, if the generic fiber of the local factor specializing to x is $\prod_j L_j$ over the fraction field F , then $e_\pi(x) = \sum_j [L_j : F]_{\text{sep}}$; see [27, §1, especially (1.2)–(1.4)].

Lemma 4.11. *For every closed point $x \in X$, $\lambda(x) = \iota e_\pi(x)$. Moreover, ι is a power of p dividing M , and every local Milnor number on the critical cover is divisible by ι .*

Proof. Let $y = \pi(x)$ and let $A = \mathcal{O}_{Y, y}^{\text{sh}}$ be the strict henselization, with fraction field F . Since π is finite flat, $X_A = X \times_Y \text{Spec } A$ is finite flat over A . Because A is strictly henselian, X_A decomposes into the local factors corresponding to the points of the geometric fiber. Let B_x be the factor corresponding to x . Its rank is the local special fiber length: $\text{rank}_A B_x = \lambda(x)$.

Since Y is normal, A is a normal local domain, and $\frac{F}{k(Y)}$ is separable algebraic. Thus separable base change preserves the inseparable degree of $\frac{k(X)}{k(Y)}$. The generic algebra $B_x \otimes_A F$ is a finite product of fields $L_1, \dots, L_s$, and every $\frac{L_j}{F}$ has inseparable degree ι . Hence $\lambda(x) = \dim_F(B_x \otimes_A F) = \sum_j [L_j : F] = \iota \sum_j [L_j : F]_{\text{sep}}$. By the preceding étale-local definition of local degree, the last sum is $e_\pi(x)$. Therefore $\lambda(x) = \iota e_\pi(x)$.

Finally $M = [k(X) : k(Y)] = \iota [k(X) : k(Y)]_{\text{sep}}$, so ι is a power of p dividing M . $\square$

Remark 4.12. The local degree used here is the algebro-geometric local number of sheets of a finite map. Gaffney–Lazarsfeld used this invariant to study higher ramification loci of finite coverings [11]; the form and semicontinuity statements used above are developed in the thesis of Lazarsfeld [27]. The factor $\iota = [k(X) : k(Y)]_{\text{insep}}$ is the extra positive-characteristic thickness that is absent for a generically separable complex covering.

For $m \geq 1$ put $\Lambda_{\geq m} = \{x \in X \mid \lambda(x) \geq m\}$. By Lemma 4.11, the local length is the constant inseparable factor ι times the local degree e_π . Since the local degree is upper semicontinuous, each $\Lambda_{\geq m}$ is closed. We will use these closed length strata to compare proper and right modality.

Proposition 4.13. *After shrinking X around x_0 , the following holds for every closed point $x \in X$. Put $h = G_x$ and $m = \mu(h) = \lambda(x) > 0$. Then*

$$\text{codim}_{X, x} \Lambda_{\geq m} = m - 1 - \text{pmod}(h).$$

Equivalently, $\dim_x \Lambda_{\geq m} = M + n - m + \text{pmod}(h)$.

Proof. Choose $N \geq 2M$ and put $E_N = \frac{\mathfrak{m}^2}{\mathfrak{m}^{N+1}}$ in the y -variables. For $v \in E_N$ choose a lift $\tilde{v}$ and set $\mu_N(v) = \dim_k \left(\frac{k[[y]]}{\text{Jac}(\tilde{v}) + \mathfrak{m}^{M+1}} \right)$. Changing the lift changes its partial derivatives

only by elements of $\mathfrak{m}^N \subseteq \mathfrak{m}^{M+1}$, so μ_N is well defined.

Suppose $\mu_N(v) = m \leq M$. The maximal ideal of the Artin algebra $\frac{k[[y]]}{\text{Jac}(\tilde{v}) + \mathfrak{m}^{M+1}}$ has m -th power zero, so $\mathfrak{m}^m \subseteq \text{Jac}(\tilde{v}) + \mathfrak{m}^{M+1}$. Put

$$L_v = \frac{\mathfrak{m}^m + \text{Jac}(\tilde{v})}{\text{Jac}(\tilde{v})}.$$

Modulo $\text{Jac}(\tilde{v})$, the inclusion $\mathfrak{m}^m \subseteq \text{Jac}(\tilde{v}) + \mathfrak{m}^{M+1}$ gives $L_v \subseteq \mathfrak{m}^{M+1-m} L_v$. Since $m \leq M$, one has $M+1-m \geq 1$, and therefore $L_v \subseteq \mathfrak{m} L_v$. Nakayama's lemma gives $L_v = 0$. Thus $\mathfrak{m}^m \subseteq \text{Jac}(\tilde{v})$, and therefore $\mathfrak{m}^{M+1} \subseteq \text{Jac}(\tilde{v})$. Consequently $\mu(\tilde{v}) = \mu_N(v) = m$. The converse is immediate. Hence $\mathcal{M}_m = \{v \in E_N \mid \mu_N(v) = m\}$ is exactly the Milnor-number- m locus and is a locally closed determinantal subset.

At h one has $T_h = \frac{\mathfrak{m} \text{Jac}(h)}{\mathfrak{m}^{N+1}} \subseteq E_N$, $\dim_k \left(\frac{E_N}{T_h} \right) = \dim_k \left(\frac{\mathfrak{m}^2}{\mathfrak{m} \text{Jac}(h)} \right) = m-1$. Choose a linear complement V to T_h . A basis of V , together with $y_1, \dots, y_n$, gives a basis of $\frac{\mathfrak{m}}{\mathfrak{m} \text{Jac}(h)}$, so $h + \sum_i a_i y_i + v$, $v \in V$, is semiuniversal with trivial section. For $m > 0$, its Milnor-number- m locus is contained in $a = 0$. Therefore $\dim_0 \{v \in V \mid \mu(h+v) = m\} = \text{pmod}(h)$.

The map $\Theta : \mathcal{R}_N \times (j^N h + V) \rightarrow E_N$, $(\varphi, j^N h + v) \mapsto \varphi(j^N h + v)$ has surjective differential at $(e, j^N h)$ because the group directions give T_h and V gives a complement. Thus Θ is smooth there. Since right equivalence preserves Milnor number, $\text{codim}_{E_N, j^N h} \mathcal{M}_m = m-1 - \text{pmod}(h)$.

Now let $\Phi : X \rightarrow E_N$, $z \mapsto j^N G_z$. In the original critical-incidence coordinates, $D_j = \frac{\partial G}{\partial t_j} = h_j(u+y) - h_j(u) - \sum_i \partial_i h_j(u) y_i$. At x_0 these classes form a basis of $\frac{\mathfrak{m}^2}{\mathfrak{m} \text{Jac}(f)}$. Consider the finite relative module $\mathcal{E}_N = \frac{\langle y_1, \dots, y_n \rangle^2}{\langle y_1, \dots, y_n \rangle \text{Jac}_y(G) + \langle y_1, \dots, y_n \rangle^{N+1}}$. The cokernel of the sections $D_1, \dots, D_{M-1}$ vanishes at x_0 and therefore vanishes after shrinking X . Every local fiber length is at most the total rank M , hence $\mu(G_z) \leq M$ and $\langle y_1, \dots, y_n \rangle^{N+1} \subseteq \langle y_1, \dots, y_n \rangle \text{Jac}(G_z)$. Thus the fiber of $\mathcal{E}_N$ at z is exactly $\frac{\mathfrak{m}^2}{\mathfrak{m} \text{Jac}(G_z)}$. It follows that the critical-family directions surject onto $\frac{E_N}{T_{G_z}}$.

Therefore $\Psi : \mathcal{R}_N \times X \rightarrow E_N$, $(\varphi, z) \mapsto \varphi(\Phi(z))$ is smooth at (e, x) . Smooth base change preserves local codimension, so $\text{codim}_{X, x} \{z \mid \mu(G_z) = m\} = m-1 - \text{pmod}(h)$. Finally λ is upper semicontinuous and $\lambda(x) = m$, so after shrinking around x there are no points with $\lambda \geq m+1$. Hence the exact- m locus agrees locally with $\Lambda_{\geq m}$. $\square$

The next step connects orbit parameters with the length strata just constructed. We start from a stratum realizing right modality and show that it gives a family of proper moduli. Let Y carry an algebraic family of germs and let $\iota_N : Y \rightarrow \mathcal{J}_N$ be its parameter-to-jet map for a sufficiently large N . We call $\{z \in Y \mid \iota_N(z) \in \mathcal{R}_N \cdot \iota_N(y)\}$ the *right-equivalence fiber* of y ; for sufficiently large N this set is independent of the chosen jet level.

Lemma 4.14. *Let Y be irreducible and suppose the family has constant Milnor number $m > 0$. If the general right-equivalence fiber has dimension i , then a general member g satisfies $\text{pmod}(g) \geq \dim Y - i$.*

Proof. Choose a general member g and a jet level $N \geq 2m$ sufficiently high for right determinacy and separability. Let $T_g = T_{j^N g}(\mathcal{R}_N \cdot j^N g)$ and choose a linear complement $\mathcal{J}_N = T_g \oplus U$. Then $\dim U = m + n - 1$. The action map $\Theta : \mathcal{R}_N \times (j^N g + U) \rightarrow \mathcal{J}_N$ is smooth at $(e, j^N g)$.

Form the fiber product, using $\iota_N : Y \rightarrow \mathcal{J}_N$ and the action map Θ , $Z = Y \times_{\mathcal{J}_N} (\mathcal{R}_N \times (j^N g + U))$ near the point lying over g . The projection $Z \rightarrow Y$ is a base change of Θ , hence smooth of relative dimension $\rho = \dim \mathcal{R}_N + \dim U - \dim \mathcal{J}_N$. Since the orbit codimension is $m + n - 1 = \dim U$, one has $\rho = \dim \text{Stab}_{\mathcal{R}_N}(j^N g)$, and this stabilizer dimension is constant on the Milnor-number- m locus.

Consider the projection $\sigma : Z \rightarrow U$. For fixed u , the image in Y of a fiber of σ lies in a single right-equivalence fiber, whose general dimension is at most i . For fixed u and a fixed member of that right-equivalence fiber, the choices of the group element form a stabilizer coset of dimension ρ . Hence the general fiber of σ has dimension at most $i + \rho$. Therefore $\dim \overline{\sigma(Z)} \geq \dim Z - (i + \rho) = \dim Y - i$.

Every $j^N g + u$ occurring in $\sigma(Z)$ is right equivalent to a member of Y , hence has Milnor number m . Shrink the slice $g + U$ around g so that $\mu \leq m$, which is possible by upper semicontinuity. On the other hand the condition $\mu \geq m$ is closed on the finite jet space (it is a determinantal rank condition), so the closure of $\sigma(Z)$ in this neighborhood is still m -constant. It contains $0 \in U$ in its closure. Consequently $\text{pmod}(g) \geq \dim \overline{\sigma(Z)} \geq \dim Y - i$. $\square$

Lemma 4.15 (Higher purity for local degree). *Let $\pi : X \rightarrow Y$ be a finite surjective morphism of smooth integral varieties and let $x_0 \in X$. Write $d = e_\pi(x_0)$. After replacing X and Y by sufficiently small neighborhoods of x_0 and $\pi(x_0)$, the following holds. If $T \subseteq X$ is an irreducible subvariety through x_0 whose generic local degree $e_\pi(T)$ is strictly less than d , then there is an irreducible codimension-one subvariety $T_1 \subseteq T$ through x_0 with*

$$e_\pi(T_1) > e_\pi(T).$$

Consequently, if $e_\pi(T) \geq e_0$, then there is an irreducible $T^ \subseteq T$ through x_0 with generic local degree d and*

$$\text{codim}_T(T^*) \leq d - e_0.$$

Proof. The codimension-one increase is the local form of the higher-purity theorem of Lazarsfeld [27, Theorem 2.2 and statement (2.4)]. Iterating it, the generic local degree increases by at least one at each step and can never exceed d . Starting from a generic degree at least e_0 , at most $d - e_0$ steps are required. $\square$

Theorem 4.16. *For every isolated hypersurface singularity over an algebraically closed field of characteristic $p > 0$,*

$$\text{pmod}(f) = \text{Rmod}(f).$$

Proof. Theorem 4.10 gives $\text{pmod}(f) \leq \text{Rmod}(f)$. Put $r = \text{Rmod}(f)$ and assume $r > 0$.

Use the semiuniversal unfolding $F(x; a, t) = H(x, t) + \sum_i a_i x_i$ introduced above. By the Rosenlicht parameter-count description of modality, there is an irreducible locally closed subset Y_0 of the parameter space, with $0 \in \overline{Y_0}$, such that the general right-equivalence fiber on Y_0 has dimension i and $\dim Y_0 - i = r$. Replacing Y_0 by a dense open subset, we may suppose that its Milnor number is constant, say m . Since $r > 0$, one must have $m > 0$. This is true since $\dim Y_0 - i = r > 0$, the general member of Y_0 cannot be smooth, because all smooth germs form a single right-equivalence class. Thus $m > 0$. In the chosen semiuniversal coordinates, $\frac{\partial F}{\partial x_i}(0; a, t) = a_i$. Hence every parameter value with $a \neq 0$ gives a smooth germ. The dense Milnor-number- m part of Y_0 is therefore contained in the linear subspace $a = 0$; after replacing Y_0 by this dense open subset, we may assume $a = 0$ identically on Y_0 .

By Lemma 4.14, for a general member g of Y_0 , $\text{pmod}(g) \geq r$.

Because $a = 0$, the point $u = 0$ is a critical point of every member of Y_0 . Thus the dense Milnor-number- m part of Y_0 has a canonical lift to the critical incidence space. Pull this lifted set through the étale localization defining X , and choose an irreducible component Z whose closure contains x_0 . For general $z \in Z$, $\lambda(z) = m$, $\text{pmod}(G_z) \geq r$. Put $D = M + n - m + r$. Proposition 4.13 gives $\dim_z \Lambda_{\geq m} \geq D$ for general $z \in Z$.

After shrinking around x_0 , write $\Lambda_{\geq m} = S_1 \cup \cdots \cup S_N$ as its irreducible decomposition. The union of those S_j of dimension at least D contains a dense open subset of Z , hence contains Z . Since Z is irreducible, it lies in one such component, say S . Thus $x_0 \in S$, $\dim S \geq M + n - m + r$.

Let $\iota = [k(X) : k(Y)]_{\text{insep}}$. By Lemma 4.11, $e_\pi(x_0) = \frac{M}{\iota}$, $e_\pi(S) \geq \frac{m}{\iota}$. Since $e_\pi(S) \geq \frac{m}{\iota}$ and $e_\pi(x_0) = \frac{M}{\iota}$, Lemma 4.15 produces an irreducible subvariety $S^* \subseteq S$ through x_0 with generic local degree $\frac{M}{\iota}$ and

$$\text{codim}_S(S^*) \leq \frac{M - m}{\iota}.$$

By Lemma 4.11, the generic local length on S^* is therefore M . Since $\Lambda_{\geq M}$ is closed, $S^* \subseteq \Lambda_{\geq M}$, and

$$\dim S^* \geq M + n - m + r - \frac{M - m}{\iota}.$$

Applying Proposition 4.13 at x_0 gives $\dim_{x_0} \Lambda_{\geq M} = n + \text{pmod}(f)$. Consequently $\text{pmod}(f) \geq r + (M - m) \left(1 - \frac{1}{\iota}\right) \geq r = \text{Rmod}(f)$. Together with the reverse inequality already proved, this gives $\text{pmod}(f) = \text{Rmod}(f)$. $\square$

Remark 4.17. Gabrielov proved equality of modality and proper modality for isolated complex analytic hypersurface singularities [12]. In positive characteristic, Nguyen proved the equality for univariate germs [32] and for all germs of right modality at most two [33];

in the latter paper the general equality was formulated as Conjecture 2.2. Theorem 4.16 proves that conjecture for arbitrary isolated hypersurface singularities.

Corollary 4.18. *With ι as above,*

$$\text{pmod}(f) = \text{Rmod}(f) \geq \mu(f) \left(1 - \frac{1}{\iota}\right).$$

If $p \nmid \mu(f)$, then $\iota = 1$. If $p = 2$ and n is odd, then $\iota \geq 2$ and

$$\text{pmod}(f) = \text{Rmod}(f) \geq \frac{\mu(f)}{2}.$$

Proof. Since $\lambda = \iota e_\pi$ and $M = \mu(f)$, $\Lambda_{\geq M} = R_{\frac{M}{\iota}-1}(\pi)$ in Lazarsfeld notation. His Theorem 2.2 gives $\text{codim}_{X,x_0} \Lambda_{\geq M} \leq \frac{M}{\iota} - 1$. Proposition 4.13 gives $\text{codim}_{X,x_0} \Lambda_{\geq M} = M - 1 - \text{pmod}(f)$, which proves the inequality. Since ι is a power of p dividing M , the condition $p \nmid M$ forces $\iota = 1$. In characteristic 2, the Hessian in the x -variables has zero diagonal and is alternating. The Jacobian matrix of the critical map Π is block triangular, with this Hessian (up to sign) as its u -block and an identity matrix in the t -block. If n is odd, the Hessian determinant vanishes identically, so $d\Pi$ is not generically of full rank. Since Π is finite dominant, it is generically inseparable and $\iota \geq 2$. $\square$

Corollary 4.19. *Assume $p = 2$ and let $c = c(f)$ be odd. Then*

$$\text{Rmod}(f) \geq \frac{\mu(f)}{2} \geq 2^{c-1}.$$

Proof. Split off the nonsingular quadratic part and let g be the residual germ in c variables. Suspension invariance gives $\mu(g) = \mu(f)$ and $\text{Rmod}(g) = \text{Rmod}(f)$. Since c is odd, Corollary 4.18 applied to g gives $\text{Rmod}(g) \geq \frac{\mu(g)}{2}$. For a residual germ in characteristic two one has $\text{Jac}(g) \subseteq \mathfrak{m}_x^2$. The Jacobian ideal is an $\mathfrak{m}_x$ -primary parameter ideal in the regular local ring $k[[x_1, \dots, x_c]]$; hence its colength equals its Hilbert–Samuel multiplicity. Monotonicity of multiplicity under inclusion gives $\mu(g) = e(\text{Jac}(g)) \geq e(\mathfrak{m}_x^2) = 2^c$. Combining the two inequalities proves the claim. $\square$

Combining the Frobenius lower bound, the comparison with proper modality, and the equality of proper and right modality summarizes the two lower-bound mechanisms used above. The first comes from deformation directions surviving modulo the right tangent image, while the second comes from the Frobenius length estimate:

$$\text{Fmod}(f) \leq d_0(f) \leq \text{pmod}(f) = \text{Rmod}(f), \quad \text{pmod}(f) \geq \left\lceil \frac{\mu(f) + n}{p^n} \right\rceil - 1.$$

Proposition 4.20. *Every isolated hypersurface germ satisfies $\text{pmod}(f) = \text{Rmod}(f) \leq \mu(f) - 1$.*

Proof. Choose a basis $x_1, \dots, x_n, h_1, \dots, h_{\mu-1}$ of $\frac{\mathfrak{m}}{\mathfrak{m} \text{Jac}(f)}$ with $h_j \in \mathfrak{m}^2$. A nonzero linear parameter in the corresponding semiuniversal unfolding makes the germ smooth, and all smooth germs form one right orbit. Hence every positive-dimensional quotient is contained in the $(\mu - 1)$ -dimensional subspace on which the linear parameters vanish. $\square$

Remark 4.21. If $n = 1$, then we have that $\text{Fmod}(f) = \text{pmod}(f) = \text{Rmod}(f) = \left\lfloor \frac{\mu(f)}{p} \right\rfloor$. This is true since the right-modality formula is the univariate classification of Nguyen [32] and the Frobenius formula follows from the fact that $\mathfrak{m} \text{Jac}(f) = \langle x^{\mu+1} \rangle$, and Theorem 4.16 identifies proper and right modality.

5 Finite classification at bounded modality

The goal of this section is to show that, after nonsingular quadratic suspension, bounded right modality gives a finite algebraic parametrization. More precisely, we construct finitely many modular families whose dimensions are bounded by the modality and whose numbers of variables and degrees are bounded explicitly in terms of p and D .

Retain the notation $C_{p,D}$, $B_{p,D}$, $N_{p,D}$, and $\widehat{N}_{p,D}$ from Section 4. For $c \geq 1$ write $\mathfrak{m}_c = \langle x_1, \dots, x_c \rangle \subseteq k[[x_1, \dots, x_c]]$; the analogous notation $\mathfrak{m}_d$ is used when a second variable count occurs. For brevity put $C = C_{p,D}$ and $M_{p,D} = p^C(D+1) - C$. For $c, L \geq 1$ write $\mathcal{H}_{c,L} = \left\{ j^L h \in \frac{\mathfrak{m}_c^2}{\mathfrak{m}_c^{L+1}} \mid \text{Hess}(h)(0) = 0 \right\}$. This is a closed, right-invariant locus of jets with no nonsingular quadratic summand left to split off. By Lemma 3.4(vii), every stable class has a residual representative in some $\mathcal{H}_{c,L}$; in characteristic 2, a square quadratic term has zero polar Hessian and is allowed in the residual part.

Definition 5.1. Let an algebraic group G act on a variety X . A locally closed subvariety $T \subseteq X$ is *orbit-finite* if every G -orbit meets T in finitely many points. Thus the parameters are genuine moduli up to a finite ambiguity; this is the finite-fiber notion usually called a *modular family* in singularity classification. A finite collection of such subvarieties is a *finite modular atlas* for an invariant constructible subset if every orbit meets at least one of them.

Lemma 5.2. *Let an algebraic group G act on a variety X , and let $W \subseteq X$ be invariant and constructible with G -modality at most D . Then W admits a finite modular atlas $T_1, \dots, T_s$ with $\dim T_i \leq D$.*

Proof. Choose a Rosenlicht stratification and reduce to a geometric quotient $q : V \rightarrow Y$ with $V \subseteq W$ and V, Y irreducible. The definition of modality gives $\dim Y \leq D$.

The generic fiber $V_{k(Y)}$ has a closed point whose residue field is finite over $k(Y)$. The closure T of that point in V is dominant and generically finite over Y . After shrinking Y it is quasi-finite and surjective, so it meets every orbit over that open set in finitely many points. Apply the same construction over the complement. Noetherian induction terminates, and repeating this on the finitely many Rosenlicht pieces proves the claim. $\square$

Lemma 5.3 (Cancellation of nonsingular quadratic suspensions). *Let $g \in \mathfrak{m}_c^2$ and $h \in \mathfrak{m}_d^2$ be residual germs, so that neither has a nonsingular quadratic summand. If g and h are stably right equivalent, then $c = d$ and $g \stackrel{\mathcal{R}}{\sim} h$.*

Proof. By stable right equivalence there are nonsingular quadratic forms Q and $\tilde{Q}$ such that $g + Q \mathcal{R} h + \tilde{Q}$. The uniqueness statement in Lemma 3.4(vii) identifies the residual parts of right-equivalent germs, also in characteristic 2; see [15]. Hence the residual variable counts agree and the residual germs are right equivalent. $\square$

Theorem 5.4. *Fix $p > 0$ and $D \geq 0$. Up to nonsingular quadratic suspension, the isolated germs f with $\text{Rmod}(f) \leq D$ are covered by finitely many algebraic families $F_i(x, t)$, $t \in T_i$, such that:*

- (i) $\dim T_i \leq D$, the number of variables is at most $C_{p,D}$, and $\deg_x F_i \leq N_{p,D}$;
- (ii) every stable right-equivalence class occurs;
- (iii) each stable class occurs at only finitely many points of each T_i .

The T_i may be refined so that Milnor number and right modality are constant.

Proof. Let $M = M_{p,D}$ and choose $L \geq \max\{2M, M + 1, N_{p,D}\}$. The residual corank $c = 0$ contributes one stable class, represented by the zero residual germ; include it as a zero-dimensional family. We may therefore assume $c \geq 1$. For each $1 \leq c \leq C_{p,D}$ let $E_{c,L} = \frac{\mathfrak{m}_c^2}{\mathfrak{m}_c^{L+1}}$ and let $\mathcal{R}_{c,L}$ be the algebraic right-jet group.

For a jet v choose a lift h and define $\mu_L(v)$ as the dimension of $\frac{k[[x_1, \dots, x_c]]}{\text{Jac}(h) + \mathfrak{m}_c^{M+1}}$. This is independent of the lift because $L \geq M + 1$. The condition $\mu_L(v) \leq M$ is determinantal and hence constructible. If $\mu_L(v) = m \leq M$, the maximal ideal of the length- m truncated Jacobian algebra has m -th power zero, so $\mathfrak{m}_c^m \subseteq \text{Jac}(h) + \mathfrak{m}_c^{M+1}$. The Nakayama lemma upgrades this to $\mathfrak{m}_c^m \subseteq \text{Jac}(h)$; hence the truncated length equals the actual Milnor number. The converse is immediate.

Intersect with the closed locus $\mathcal{H}_{c,L}$. Since $L \geq 2M$, every isolated germ there is right L -determined, so its formal right modality is the modality of its L -jet. Upper semicontinuity of modality for algebraic group actions [14, Proposition 2.7] makes the locus $W_{c,D}$ of modality at most D invariant and constructible.

Theorem 4.3 shows that every stable class with modality at most D has a residual representative in one of the finitely many $W_{c,D}$. Apply Lemma 5.2. The tautological L -jets over the resulting orbit-finite multisections are algebraic polynomial families. Since every fiber is right $N_{p,D}$ -determined, truncate to degree $N_{p,D}$. This does not change the represented formal orbit. It also preserves orbit-finiteness: if two truncated parameters lie in the same $N_{p,D}$ -jet orbit, their corresponding formal germs are right equivalent by $N_{p,D}$ -determinacy. Conversely, because all representatives under consideration are residual, Lemma 5.3 shows that two of them represent the same stable class only if they are right equivalent. Thus each stable class occurs at only finitely many parameters on each multisection. Finally refine by the finitely many Milnor numbers and integer modality values. $\square$

Remark 5.5. Greuel–Nguyen observed that, for each fixed positive characteristic, only finitely many right-simple singularities occur and conjectured an analogous finiteness phenomenon for an arbitrary number of moduli [14]. Nguyen subsequently gave explicit finite lists of modular normal-form families in modalities one and two [33]. Theorem 5.4 gives a stable, finite-atlas form of this finiteness phenomenon in every modality. Thus it also realizes the general finiteness expectation stated in the introduction of [14]. It is a finite algebraic parametrization statement rather than an explicit list of normal forms. For a complementary ADE-type classification over complete regular local rings, organized by Cohen–Macaulay type rather than right modality, see [39].

Corollary 5.6. *For every fixed characteristic and finite modality bound, stable right classification is a finite algebraic classification problem: finitely many modular families of dimension at most the modality suffice. In modality zero this gives finiteness of stable right-simple classes without using an explicit normal-form classification.*

6 Applications to F -singularities

In this section, we apply the classification results of the previous section to F -singularity invariants. We first recall the definitions that we use, and then prove uniform results for the F -pure threshold, F -jumping numbers, test ideals, and positive-characteristic Bernstein–Sato roots.

Throughout this section k is algebraically closed of characteristic $p > 0$. A ring A of characteristic p is F -finite if the Frobenius pushforward F_*A is a finite A -module. Here $F_*^e A$ denotes the additive group of A , regarded as an A -module by $a \cdot F_*^e b = F_*^e(a^{p^e}b)$. Since k is perfect, the regular power-series ring R is F -finite.

For $e \geq 0$ put $q = p^e$. For an ideal $I \subseteq R$, its q -th Frobenius power is $I^{[q]} = \langle a^q \mid a \in I \rangle$; in particular $\mathfrak{m}^{[q]} = \langle x_1^q, \dots, x_n^q \rangle$. We write $\nu_e(f) = \max\{a \in \mathbb{N} \mid f^a \notin \mathfrak{m}^{[p^e]}\}$. The F -pure threshold of the principal ideal $\langle f \rangle$ at $\mathfrak{m}$ is $\text{fpt}(f) = \lim_{e \rightarrow \infty} \frac{\nu_e(f)}{p^e}$; equivalently, it is the first positive jumping number of the generalized test-ideal filtration.

The q -th Frobenius root $I^{[\frac{1}{q}]}$ is the unique smallest ideal J such that $I \subseteq J^{[q]}$. In a regular F -finite ring the generalized test ideal of a principal ideal is, for all sufficiently large e , $\tau(f^t) = \langle f^{[tp^e]} \rangle^{[\frac{1}{p^e}]}$; the positive real numbers at which $\tau(f^t)$ changes are the F -jumping numbers of $\langle f \rangle$ [4]. We call their set in a specified interval the *numerical F -jumping spectrum*; this forgets the test ideals themselves and remembers only the jumping numbers. A homomorphism of A -modules is *pure* if it remains injective after tensoring with every A -module. A reduced F -finite ring A is F -pure if the Frobenius map $A \rightarrow F_*A$ is pure (equivalently, for an F -finite ring, split), and it is *strongly F -regular* if for every c outside the union of the minimal primes some A -linear map $F_*^e A \rightarrow A$ sends $F_*^e c$ to 1 [23]. Finally, $\tau_{\text{fg}}(A)$ denotes the *finitistic test ideal*, the intersection of $\text{Ann}_A(0_M^*)$ over all finitely generated A -modules M . Here 0_M^* is the *tight closure of zero*: an element $z \in M$ lies in 0_M^* if there is $c \in A$ outside the union of the minimal primes such that $cF_M^e(z) = 0$.

in $F_A^e(M)$ for all sufficiently large e , where $F_A^e(M)$ denotes the e -fold Frobenius functor applied to M . These are the conventions used in the sources cited below.

Remark 6.1. The Fedder criterion made Frobenius splitting of hypersurfaces computable [10]. Hara–Yoshida introduced generalized test ideals [20], while Mustață–Takagi–Watanabe introduced and studied F -thresholds and related them to the jumping behavior of test ideals [31]. Blickle–Mustață–Smith proved discreteness and rationality in the regular F -finite setting [4]. Mustață then constructed positive-characteristic Bernstein–Sato polynomials and related their roots to F -jumping numbers [30]; Blickle–Stäbler extended this framework to test modules [5].

Let f be isolated, put $r = \text{Rmod}(f)$, and write $f \stackrel{\mathcal{R}}{\sim} g(x_1, \dots, x_c) + Q$ as in Lemma 3.4(vii), with $c = c(f)$. By Proposition 4.1, the Frobenius Loewy length is unchanged by nonsingular quadratic suspension. Therefore the residual quantity $B_F^{\text{res}} = p(\ell_F(f) - 1) + c(p - 1)$ is exactly the bound $B_F(g)$ for the residual germ. We use this quantity in the following threshold estimate.

Theorem 6.2. (i) $\text{fpt}(f) \geq \frac{1}{\max\{2, 2B_F^{\text{res}}\}}$.

(ii) If $c > 0$, then $\text{fpt}(f) \geq \frac{1}{2B_F^{\text{res}}} \geq \frac{1}{2pr + 2c(p - 1)}$.

(iii) If $c = 0$, then $\text{fpt}(f) = \frac{1}{2}$ in one variable and odd characteristic; if the quadratic form contains a hyperbolic plane, then $\text{fpt}(f) = 1$.

Proof. Apply Theorem 3.25 to the residual germ g . Since $\ell_F(g) = \ell_F(f)$ by Proposition 4.1, a right coordinate change in the x -variables sends g to a polynomial $g_{\leq 2B_F^{\text{res}}}$ of degree at most $2B_F^{\text{res}}$. Extending that coordinate change trivially across the variables of Q shows that f is right equivalent to $g_{\leq 2B_F^{\text{res}}} + Q$, whose degree is at most $\max\{2, 2B_F^{\text{res}}\}$.

If h is a nonzero polynomial of degree at most N , put $a = \left\lfloor \frac{q-1}{N} \right\rfloor$ for $q = p^e$. Then h^a is nonzero and has total degree less than q , hence cannot lie in $\langle x_1^q, \dots, x_n^q \rangle$. Therefore $\nu_e(h) \geq a$, and passage to the limit gives $\text{fpt}(h) \geq \frac{1}{N}$. This proves the first bound. If $c > 0$, then $B_F^{\text{res}} \geq c(p - 1) \geq 1$, so the maximum is $2B_F^{\text{res}}$; moreover $\ell_F(f) - 1 \leq \text{Fmod}(f) \leq r$, which gives the second bound.

If $c = 0$, Lemma 3.4(vii) leaves only the nonsingular quadratic form. Over an algebraically closed field, a nonsingular quadratic form in at least two variables contains a hyperbolic plane uv . Then Q^{q-1} contains the monomial $u^{q-1}v^{q-1}$ with nonzero coefficient, so $\nu_e(Q) = q - 1$ and $\text{fpt}(Q) = 1$. The only remaining case is one variable in odd characteristic, where Q is right equivalent to x^2 and $\text{fpt}(Q) = \frac{1}{2}$. $\square$

Proposition 6.3. If $c = c(f)$, then

$$\text{fpt}(f) \geq \frac{1 - \frac{1}{\text{ord}(f)}}{p^c(\text{Fmod}(f) + 1) - c}.$$

Proof. By [41, Corollary 2.14], $\text{fpt}(f) \geq \frac{1}{\text{Tj}(f)} \left(1 - \frac{1}{\text{ord}(f)}\right)$. Since $\text{Tj}(f) \leq \mu(f)$, apply (7). $\square$

Proposition 6.4. *Assume $p \nmid N$ and let $f_N = xy - z^N$. Then $\frac{k[[x,y,z]]}{\langle f_N \rangle}$ is strongly F -regular and $\text{fpt}(f_N) = 1$, whereas $\text{Fmod}(f_N) = \text{pmod}(f_N) = \text{Rmod}(f_N) = \left\lfloor \frac{N-1}{p} \right\rfloor$.*

Proof. The hyperbolic block xy splits off, leaving z^N . The univariate formula and Theorem 4.16 give the modality statement. The ring is the tame cyclic quotient $k[[u,v]]^{\mu_N}$ for the action $\zeta \cdot (u,v) = (\zeta u, \zeta^{-1}v)$, with $x = u^N$, $y = v^N$, and $z = uv$. Since $p \nmid N$, the invariant ring is a direct summand of the regular ring and hence strongly F -regular by [23]. It is F -pure, so the principal F -pure threshold is one. $\square$

Corollary 6.5. *Fix $p > 0$ and $D \geq 0$. Let f be isolated with $\text{Rmod}(f) \leq D$ and $h \in \mathfrak{m}^{\widehat{N}_{p,D}+1}$. Then:*

- (i) $f + h \stackrel{\mathcal{R}}{\sim} f$ and $\text{fpt}(f + h) = \text{fpt}(f)$;
- (ii) $f + h$ has the same numerical F -jumping spectrum as f ;
- (iii) for some local coordinate automorphism φ we have that $\tau((f + h)^\lambda) = \varphi(\tau(f^\lambda))$ for every $\lambda \geq 0$.

Proof. The first assertion is precisely Corollary 4.4. The F -pure threshold and the set of F -jumping numbers are invariant under local coordinate changes. Generalized test ideals commute with local isomorphisms, localization, and completion in the regular F -finite setting [4, Proposition 2.13]; in particular they are functorial under ring automorphisms, which gives the displayed identity. $\square$

Remark 6.6. Hernández–Núñez–Betancourt–Witt proved local $\mathfrak{m}$ -adic constancy of the F -pure threshold and, with a deeper perturbation order, literal equality of all test ideals $\tau(f^\lambda)$ for $0 \leq \lambda < 1$ around a fixed polynomial with an isolated singularity [21]. Their explicit radii depend on the Jacobian colength and the ambient dimension. Corollary 6.5 has a different form: its radius is uniform on the whole bounded-modality locus and is independent of ambient dimension, while the test ideals are identified up to the coordinate change supplied by right determinacy.

Set $C = C_{p,D}$ and $d_{p,D} = \widehat{N}_{p,D}$. Put $\delta_p = 0$ if $p = 2$ and $\delta_p = 1$ if $p > 2$, and set $n_{p,D} = C + \delta_p$. Let e_0 be the least integer with $p^{e_0} > d_{p,D}$ and set $H_{p,D} = \binom{d_{p,D} + n_{p,D}}{n_{p,D}}$. Define

$$\Xi_{p,D} = \bigcup_{\substack{0 \leq a \leq e_0 + H_{p,D} \\ 1 \leq b \leq H_{p,D}}} \left\{ \frac{j}{p^a(p^b - 1)} : 1 \leq j \leq p^a(p^b - 1) \right\}.$$

Theorem 6.7. *If f is isolated and $\text{Rmod}(f) \leq D$, then every F -jumping number of $\langle f \rangle$ in $(0, 1]$ belongs to the finite set $\Xi_{p,D}$. Consequently only finitely many numerical F -jumping spectra, and in particular only finitely many F -pure thresholds, can occur at fixed p, D , independently of the original ambient dimension.*

Proof. After splitting, either the quadratic block contains a hyperbolic plane or it has rank at most one. Over an algebraically closed field the latter alternative can occur only in odd characteristic. If the quadratic block contains no hyperbolic plane, then it is zero in characteristic two and has dimension at most one in odd characteristic. Hence, in the non-hyperbolic case, Theorem 5.4 reduces the germ to a principal polynomial in at most $n_{p,D} = C + \delta_p$ variables and of degree at most $d_{p,D}$. Test ideals and their jumping numbers are unchanged by the local coordinate changes used here and commute with localization and completion in this regular F -finite setting; see [4, Proposition 2.13]. Proposition 3.8 of [4], with one generator, therefore puts every global F -jumping exponent of this polynomial in the displayed finite denominator set. A local jump at the maximal ideal is necessarily a global jump: if the two adjacent test ideals become distinct after localization at the maximal ideal, then they were already distinct before localization. Hence every local F -jumping number occurring here belongs to the same finite denominator set.

If the quadratic block has rank at least two, it contains a hyperbolic plane uv ; by Lemma 3.4(vii) the residual term h involves variables disjoint from u, v . For every $q = p^e$, the monomial $u^{q-1}v^{q-1}$ occurs with coefficient one in $(h + uv)^{q-1}$ and cannot cancel. Thus the hypersurface is F -pure and its first jump is 1; there are no jumps in $(0, 1)$. $\square$

All reduced denominators occurring in the finite candidate set $\Xi_{p,D}$ are bounded by the single integer $Q_{p,D} = p^{e_0 + H_{p,D}}(p^{H_{p,D}} - 1)$. This gives a uniform separation between distinct candidates and will allow one Frobenius approximation to determine the F -pure threshold.

Corollary 6.8. *Any two distinct F -jumping numbers in $(0, 1]$ arising at modality at most D differ by at least $\frac{1}{Q_{p,D}^2}$.*

Proof. Every reduced denominator is at most $Q_{p,D}$. Two distinct reduced fractions with denominators at most $Q_{p,D}$ differ by at least $\frac{1}{Q_{p,D}^2}$. $\square$

Theorem 6.9. *Choose $E_{p,D}$ so that $q_{p,D} = p^{E_{p,D}} > Q_{p,D}^2 + 1$. Then the single integer $\nu_{E_{p,D}}(f)$ determines $\text{fpt}(f)$ for every isolated germ with $\text{Rmod}(f) \leq D$.*

Proof. For every principal germ and every $e \geq 0$ one has the elementary interval $\frac{\nu_e(f)}{p^e} \leq \text{fpt}(f) \leq \frac{\nu_e(f)+1}{p^e}$. Indeed, the left inequality follows from the defining Frobenius sequence, while $f^{\nu_e(f)+1} \in \mathfrak{m}^{[p^e]}$ and Frobenius iteration give the right inequality. At $e = E_{p,D}$ the interval has length $\frac{1}{q_{p,D}}$, which is strictly smaller than the separation in Corollary 6.8. Theorem 6.7 therefore leaves exactly one possible F -pure threshold. $\square$

The assertions about F -pure thresholds concern nonzero equation families. The zero residual class is represented by x^2 in odd characteristic and by uv in every characteristic. When recording the numerical invariants of suspensions, we also consider one-variable quadratic suspensions in odd characteristic and hyperbolic suspensions. Every family containing a hyperbolic plane has F -pure threshold 1.

Theorem 6.10. *The modular atlas of Theorem 5.4 can be refined into finitely many locally closed parameter spaces on which Milnor number, right modality, and F -pure threshold are all constant. On every irreducible modular family the F -pure threshold is constant on a dense open subset.*

Proof. Fix $e = E_{p,D}$ and $q = p^e = q_{p,D}$. For a polynomial family $F(x, t)$, the condition $F_t^r \in \mathfrak{m}^{[q]}$ is closed because it is the vanishing of finitely many coefficient functions after reduction modulo $\langle x_1^q, \dots, x_n^q \rangle$. Hence $\nu_e(F_t) = r$ is locally closed. Theorem 6.9 converts these finitely many strata into constant- F -pure-threshold strata. On an irreducible parameter space, one of the finitely many locally closed strata contains the generic point and therefore contains a dense open subset; this gives the final assertion. $\square$

Corollary 6.11. *Let $A = \frac{R}{\langle f \rangle}$ be reduced, write $\mathfrak{m}_A = \frac{\mathfrak{m}}{\langle f \rangle}$, and assume $\text{Rmod}(f) \leq D$. Then:*

- (i) *the number of F -jumping coefficients in $[0, 1]$ is at most $M_{p,D} + 1$;*
- (ii) $\text{length}_k \left(\frac{A}{\tau_{\text{fg}}(A)} \right) \leq M_{p,D}$;
- (iii) *if $E = pD + C_{p,D}(p - 1) + 1$, then $\mathfrak{m}_A^E \subseteq \text{Jac}(f)A \subseteq \tau_{\text{fg}}(A)$.*

Proof. Katzman–Lyubeznik–Zhang [25] bound the number of jumps by $\text{Tj}(f) + 1$. Since $\text{Tj}(f) \leq \mu(f) \leq M_{p,D}$, the first assertion follows. Because A is a reduced hypersurface complete local algebra over the algebraically closed field k , it is absolutely reduced and equidimensional. Jiang [24, Theorem 4.1] therefore gives $\text{Jac}(f)A \subseteq \tau_{\text{fg}}(A)$. Lemma 3.24, applied to the residual germ as in Proposition 4.1, gives the displayed power containment; quadratic variables already lie in the Jacobian ideal. Finally,

$$\frac{A}{\text{Jac}(f)A} \cong \frac{R}{\langle f \rangle + \text{Jac}(f)}$$

is the Tjurina algebra, and $\text{Jac}(f)A \subseteq \tau_{\text{fg}}(A)$ shows that $\frac{A}{\tau_{\text{fg}}(A)}$ is its quotient. $\square$

Following Blickle–Stäbler, for each Frobenius level e there is a finite set of positive-characteristic Bernstein–Sato roots attached to $\langle f \rangle$. The corollary below concerns only its eventual level-by-level behavior, as stated explicitly:

Corollary 6.12. *At fixed p, D , only finitely many eventual sequences of level- e positive-characteristic Bernstein–Sato root sets occur for isolated hypersurfaces of right modality at most D .*

Proof. By Theorem 6.7, only finitely many numerical F -jumping spectra in $(0, 1]$ occur at fixed p, D . In the non-hyperbolic case, the proof of that theorem replaces the germ, after splitting off its nonsingular quadratic part, by a bounded-degree polynomial in a uniformly bounded number of variables whose localization and completion have the same numerical F -jumping spectrum. In the hyperbolic case the spectrum in $(0, 1]$ is already fixed. By compatibility of generalized test ideals with completion [4, Proposition 2.13], we may therefore apply Blickle–Stäbler to these polynomial representatives. Their Theorem 5.4 gives,

for all sufficiently large e , the root set $\left\{ \frac{[\lambda p^e]-1}{p^e} : \lambda \text{ is an } F\text{-jumping number in } (0, 1] \right\}$. Since only finitely many jumping spectra occur, only finitely many eventual level-by-level Bernstein–Sato root sequences occur. $\square$

For matrices, prescribed-jet transversality reduces questions about formal equivalence to the geometry of a finite stratification by rank. For hypersurfaces, the Frobenius algebra and the critical cover reduce questions about right equivalence and modality to finite-dimensional algebraic data. In particular, once the characteristic and the modality are bounded, the remaining classification problem and the F -singularity invariants considered above are controlled by finitely many algebraic families.

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EINSTEIN INSTITUTE OF MATHEMATICS, THE HEBREW UNIVERSITY OF JERUSALEM, EDMOND J. SAFRA CAMPUS, GIV'AT RAM, JERUSALEM, 91904, ISRAEL

E-mail address: yotam.svoray@huji.mail.ac.il